8

Optical Spectroscopes

8.1 INTRODUCTION

Detectors operating in the optical part of the spectrum are generally broad-band devices, sensitive to radiation over a wide range of wavelengths. The spectrum cannot therefore be observed directly as it may at shorter wavelengths where the detector's response varies with the energy of the photon. In the optical region a separate device is required to separate out the radiation into its component wavelengths before it is detected. That device is called a Spectroscope†.

There are several processes whereby the radiation can be separated into its component wavelengths. The simplest is to use a range of filters placed before the detector to isolate particular regions of the spectrum. Traditionally, however, this method of studying the spectrum of an astronomical object is regarded as a separate subject from spectroscopy, and is called Photometry. The spectral resolution, R, is defined as

$$R = \frac{\lambda}{\Delta\lambda} \tag{8.1}$$

where λ is the operating wavelength and $\Delta\lambda$ is the smallest wavelength interval that may be resolved. With this definition, photometry is concerned with spectral resolutions of less than about 100, spectroscopy with spectral resolutions greater than about 100. This is an arbitrary division, however, and exceptions occur. Thus studying the sun using an Hα filter is generally regarded as a branch of photometry even though the spectral resolution may be around 10 000, while identifying quasars on an objective prism plate (see later) at a spectral resolution of around 50 would belong to spectroscopy. Here the separation is therefore made by regarding spectral studies employing only filters not to be a part of spectroscopy.

† Strictly this term should be reserved for visual devices, and Spectrograph used when photography is used to detect the spectrum. Spectrometer should then be used for devices using other types of detectors such as CCDs. I prefer to use the single term, spectroscope, for these devices irrespective of the detector they employ on the grounds that this is the practice applied to the term Telescope, and we do not call it a Telegraph when a camera is attached in place of an eyepiece!

Copyright © 1995 IOP Publishing Ltd.

A second method of obtaining a spectrum is based upon differential refraction and results in rainbows (chapter 1) and prism-based spectroscopes. The latter are still used in astronomy and are briefly considered later in this chapter. Most modern astronomical spectroscopes, however, utilize interference effects in some way to produce the spectrum. The resulting devices which have found application in astronomy are Diffraction Gratings, Fabry–Pérot Etalons and Fourier Transform Spectroscopes, and we consider each of these below.

8.2 DIFFRACTION GRATINGS

The effect of a diffraction grating is nowadays familiar to most people from the colours reflected off compact discs. These, like diffraction gratings, have numerous closely spaced parallel apertures, and the colours result from constructive and destructive interference between light reflected from those different apertures. Although a grating will usually contain many thousands of apertures, the principle of its operation is based upon the interference effects produced by just two apertures. Most gratings in practice are reflection gratings, but the principle of their operation is the same as that of a transmission grating. Since with the latter it is easier to visualize what is happening, the grating equation is derived for a transmission grating.

Let us consider then the Fraunhofer (far field) interference effects from a pair of closely spaced apertures (figure 8.1) illuminated by a distant monochromatic source of wavelength, λ. The path difference at an angle, θ to the incoming ray is given by (figure 8.2)

$$\Delta p = s \sin\theta \tag{8.2}$$

where s is the separation of the apertures.

The interference pattern on the screen will be a series of light and dark fringes (figure 8.3b) accordingly as the path difference is an integer, or integer plus half, number of wavelengths and we have constructive or destructive interference, respectively. The intensities of the maxima are modulated by the interference pattern from a single aperture (figure 8.3a). The centres of the bright fringes correspond to integer wavelength path differences, and so at such points we have

$$\Delta p = s \sin\theta = n\lambda \tag{8.3}$$

where n is an integer, usually referred to as the order of the interference. The angle between successive maxima is thus

$$\Delta\theta = \sin^{-1}\left(\frac{(n+1)\lambda}{s}\right) - \sin^{-1}\left(\frac{n\lambda}{s}\right) \tag{8.4}$$

which for small angles simplifies to

$$\Delta p \approx \frac{\lambda}{s}. \tag{8.5}$$

Copyright © 1995 IOP Publishing Ltd.

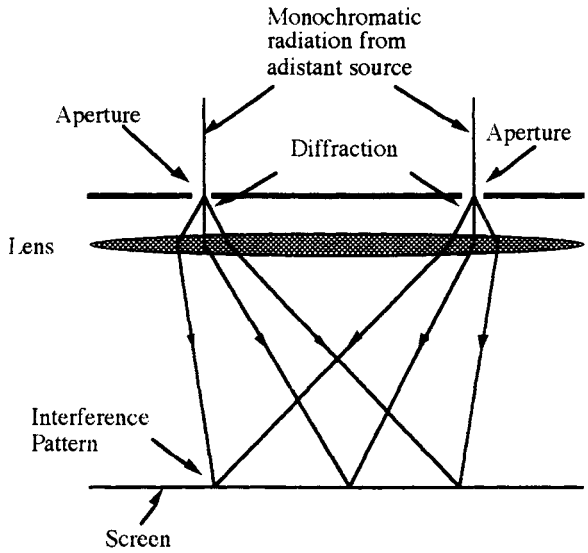

Figure 8.1 Fraunhofer interference from a pair of apertures.

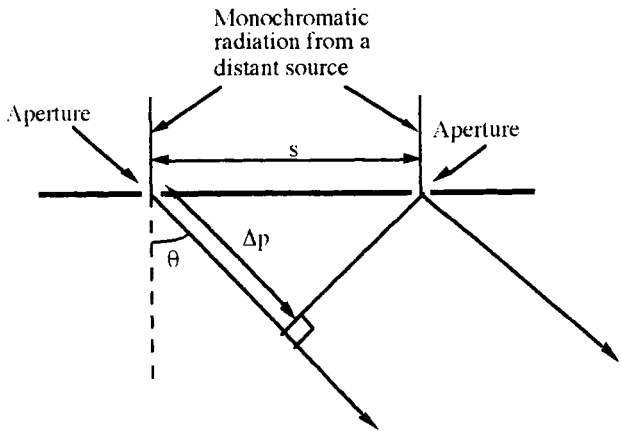

Figure 8.2 Path difference in Fraunhofer interference at two apertures.

If the normal to the plane of the apertures is at an angle f to the incoming radiation; corresponding to the often utilized inclined grating, then (figure 8.4)

$$\delta p = s \sin\theta + s \sin\phi = n\lambda \tag{8.6}$$

We then get the grating equation in its normal form

$$\theta = \sin^{-1}\left(\frac{n\lambda}{s} - \sin\phi\right) \tag{8.7}$$

Copyright © 1995 IOP Publishing Ltd.

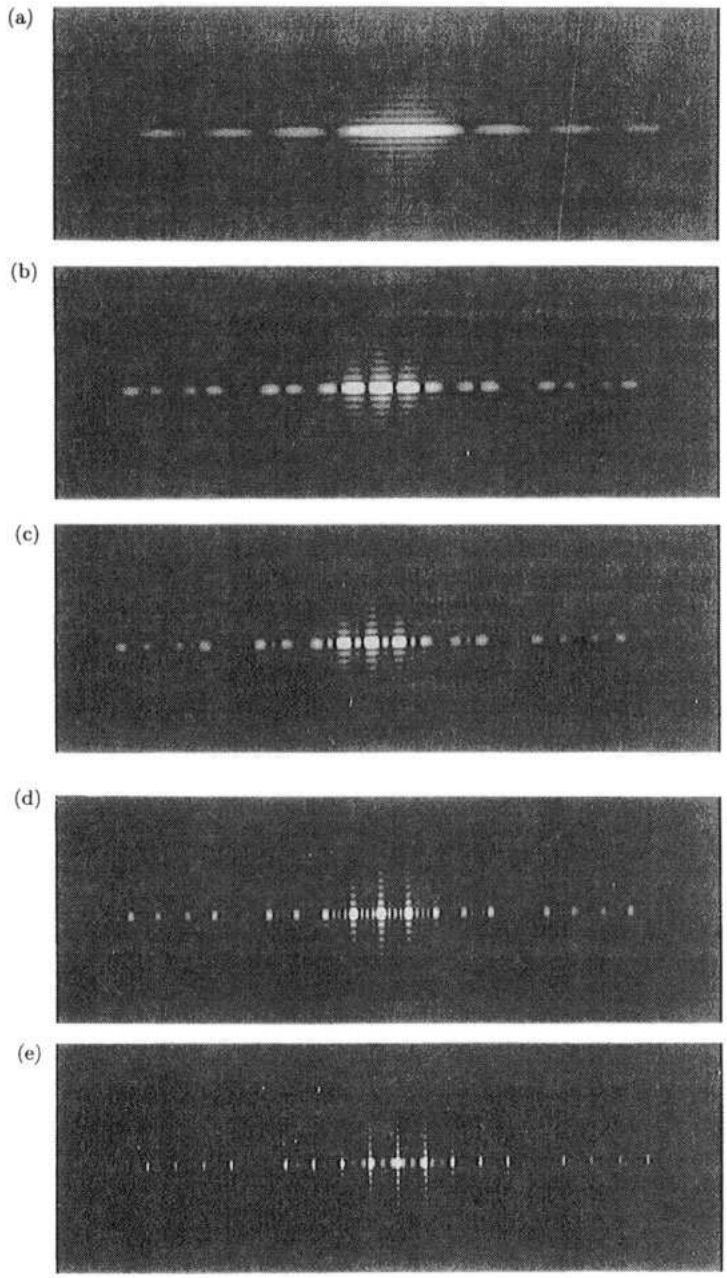

Figure 8.3 Fraunhofer interference patterns: (a) from a single aperture, (b) from two apertures, (c) from three equally spaced apertures, (d) from five equally spaced apertures, (e) from ten equally spaced apertures.

Copyright © 1995 IOP Publishing Ltd.

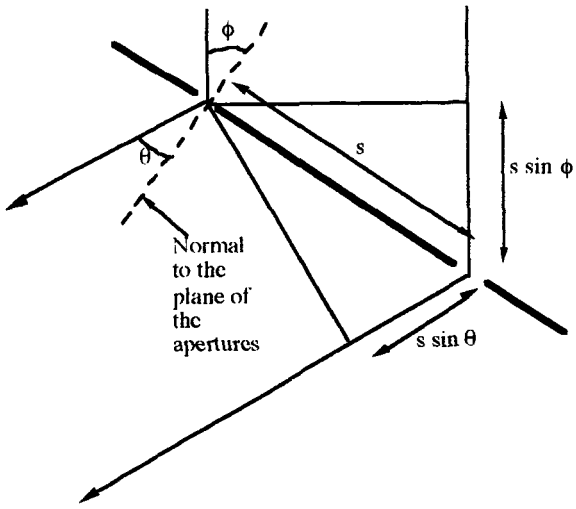

Figure 8.4 Plane of the apertures inclined to the incoming radiation.

From equation (8.7), we may see that the angular positions of the maxima are wavelength dependent. Maxima for different wavelengths will therefore appear at different angles to the incoming radiation (figure 8.5), and thus we have the principle of operation of the diffraction grating. Of course, as may also be seen from figure 8.5, the device at this stage is not much use as a practical means of producing a spectrum because the fringes are very broad and overlap. However if we add a third aperture in line with the first two and at the same spacing, then the fringe width reduces (figure 8.3c), while the separation remains constant. Weak secondary maxima also appear between the main fringes. Adding further apertures causes the primary fringe widths to reduce further (figure 8.3d), and although more secondary maxima are produced, they become progressively weaker. By the time the number of apertures is into double figures (figure 8.3e), the secondary maxima have almost disappeared and the primary maxima have become narrow enough for different wavelengths to be separated out (figure 8.6). In practice, gratings contain thousands of apertures and the angular fringe widths become equivalent to fractions of a nanometre.

8.2.1 Angular resolution

The above, almost qualitative, description of the operating principles of a diffraction grating may be made more quantitative via the intensity equation. The intensity in the interference pattern at an angle, θ, to the direction of the incoming radiation, $I(\theta)$, relative to the central intensity of the pattern, $I(0)$, is given by

$$\frac{I(\theta)}{I(0)} = \frac{\sin^2(\pi d \sin\theta/\lambda)}{(\pi d \sin\theta/\lambda)^2} \times \frac{\sin^2(N\pi s \sin\theta/\lambda)}{\sin^2(\pi s \sin\theta/\lambda)} \tag{8.8}$$

Copyright © 1995 IOP Publishing Ltd.

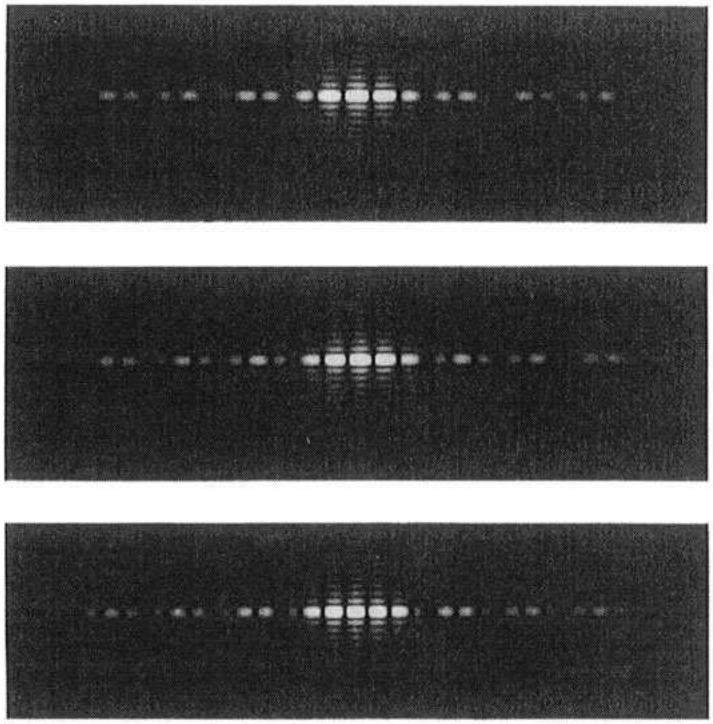

Figure 8.5 Interference patterns for monochromatic sources of different wavelengths through the same pair of apertures ($\lambda_1 > \lambda_2 > \lambda_3$).

where d is the width of an aperture and N is the number of apertures. The second fraction on the right hand side of equation (8.8) gives the widths and positions of the main fringes, and is often called the interference term. The first fraction represents a modulation of the interference pattern by the pattern for a single aperture (see figures 8.3a and 8.3e for example). Since N is an integer larger than unity, both numerator and denominator in the interference term tend to zero as $(s \sin\theta)$ tends to an integer multiple of the wavelength. The interference term as a whole, however, tends to a limit of $\pm N$. Such positions correspond to the primary maxima, as we have previously seen (equation (8.3)).

The minima on either side of a primary maximum occur as the numerator of the interference term goes to zero. Thus we have for a primary maximum of order n, occurring at an angular position of

$$\theta = \sin^{-1}\left(\frac{n\lambda}{s}\right) \tag{8.9}$$

the first zeros occurring at angular positions of

$$\theta \pm \Delta\theta = \sin^{-1}\left(\frac{(Nn \pm 1)\lambda}{Ns}\right). \tag{8.10}$$

Copyright © 1995 IOP Publishing Ltd.

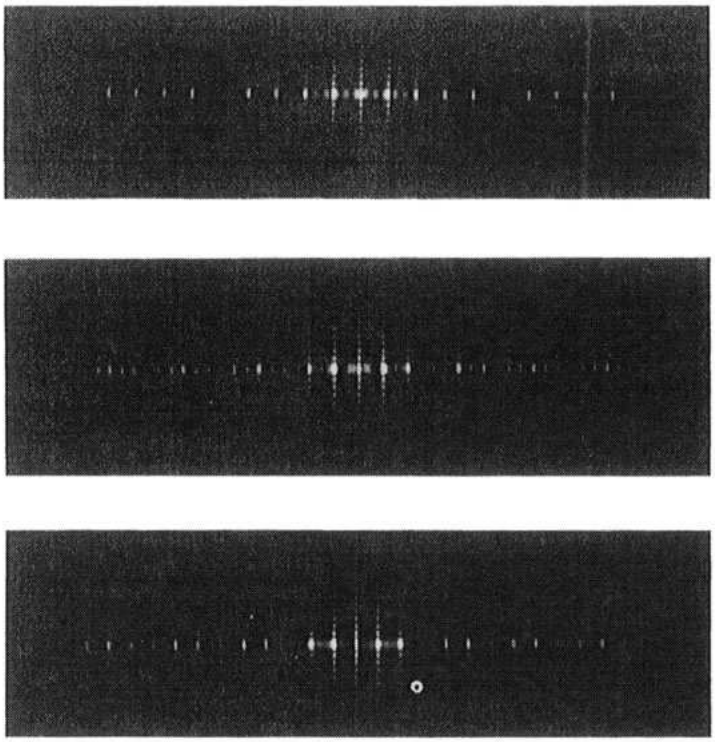

Figure 8.6 Interference patterns for monochromatic sources of different wavelengths through ten apertures ($\lambda_1 > \lambda_2 > \lambda_3$).

The angular width of a primary maximum is thus $2\Delta\theta$ and is given by

$$W = 2\Delta\theta \approx \frac{2\lambda}{Ns\cos\theta}. \tag{8.11}$$

The Rayleigh criterion for resolution is for one fringe maximum to be superimposed on the first zero of a second fringe pattern. Thus the angular Rayleigh resolution of a diffraction grating is given by $\Delta\theta$.

$$\text{Angular Rayleigh resolution} = \Delta\theta = \frac{\lambda}{Ns\cos\theta}. \tag{8.12}$$

8.2.2 Dispersion

The dispersion of a spectrum is the rate of change of wavelength with angular position. From equation (8.3) we have

$$\lambda = \frac{s\sin\theta}{n} \tag{8.13}$$

Copyright © 1995 IOP Publishing Ltd.

and so

$$\frac{d\lambda}{d\theta} = \frac{s\cos\theta}{n} \tag{8.14}$$

Since θ is generally small, the dispersion along a spectrum from a diffraction grating is approximately constant ($\cos\theta \approx 1$ for small θ). The dispersion, however, varies between differing orders, the angular spread of the spectrum being proportional to the order.

8.2.3 Spectral resolution

We have defined the spectral resolution previously (equation (8.1)), and can now calculate it for a diffraction grating. From equations (8.12) and (8.14) we have

$$\frac{\lambda}{\Delta\lambda} = R = nN \tag{8.15}$$

so that the spectral resolution of a grating spectrum depends only on the order of the spectrum and the number of apertures in the grating. For a typical astronomical spectroscope as used on a large telescope, N might be in the region of 10 000 to 20 000. The order of the spectrum would be 1 or 2, or occasionally 3. The typical spectral resolution available would therefore be in the region of 30 000, or about 0.02 nm in the visible part of the spectrum. For an echelle grating (see later in this chapter), used at order 100, the spectral resolution could reach 10^5 or more.

8.2.4 Free spectral range

The overlap between wavelengths shown in figure 8.5 is reduced to usable levels by the increase in the number of apertures in the grating. However for a real spectrum, with wavelengths extending from the radio region through to the far ultraviolet and beyond, there will still be an overlap between differing orders. The difference in wavelength between two superimposed wavelengths from adjacent orders is called the Free Spectral Range, and is usually given the symbol, Σ. If λ_1 and λ_2 are two such superimposed wavelengths, then from equation (8.9) and for small q, we have

$$\Sigma = (\lambda_1 - \lambda_2) \approx \lambda_2/n. \tag{8.16}$$

For spectroscopes operating at low spectral orders, therefore, the free spectral range is large, and any overlap can be eliminated by the use of filters. At large orders, however, Σ can reduce to a few nanometres. The overlapping orders then have to be separated by a second spectroscope whose dispersion is perpendicular to that of the first. This arrangement is called a Cross Disperser, and it is discussed further along with echelle gratings and Fabry–Pérot etalons later in this chapter.

Copyright © 1995 IOP Publishing Ltd.

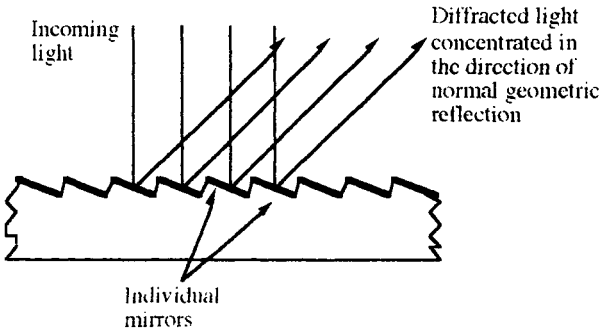

Figure 8.7 Enlarged section through a blazed reflection grating.

8.2.5 Blazing

For simplicity, we have derived the equations for diffraction gratings using apertures. The use of apertures in a real device would result in a transmission grating. In practice most gratings use mirrors in place of the apertures and result in reflection gratings. This causes no change to any of the above analysis. There are two reasons for the general preference for reflection over transmission gratings. The first is that reflection gratings are easier, and therefore cheaper, to produce (see below). The second is that reflection gratings are easier to blaze.

A major disadvantage of diffraction gratings for the production of spectra is obvious from figure 8.3e, and is that the available light is split into ten or more different orders. The spectrum that is being observed will therefore contain 10% or less of the incoming light. Since the overall intensities of the maxima are modulated by the diffraction pattern for a single aperture (figure 8.3), we may concentrate the light into a single order by widening the apertures until the central fringe of that modulation is just wider than a single spectrum. With simple apertures, however, this would concentrate the light into the zero order, and this, as we may see from equation (8.14) is not a spectrum. The solution, for reflection gratings is to tilt the individual mirrors (figure 8.7) so that the single aperture maximum is directed towards the order of spectrum that is of interest. This is called 'Blazing the grating', and can concentrate up to 90% of the incoming light into the desired spectrum. Transmission gratings can also be blazed, by replacing the aperture by a prism to refract the light into the desired spectral order.

Blazing has the further advantage of reducing the problem of overlap of different spectral orders (see above). But as may be seen from figure 8.7, it will cause some loss of light through the vertical portions of the grooves intercepting part of the reflected beam. This problem is called shadowing, and can become significant if the reflection is at a large angle to the normal to the grating.

Copyright © 1995 IOP Publishing Ltd.

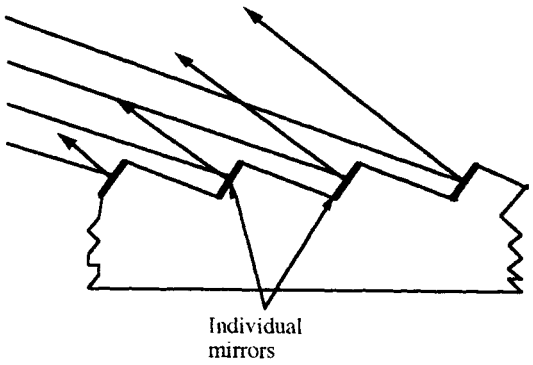

Figure 8.3 Enlarged section through an echelle grating.

8.2.6 Curved gratings

All the gratings so far considered have been plane; they must therefore have other optical components to collimate and focus the light (see later in this chapter). Those other components will cause light loss and scattering, and in the case of lenses may introduce some chromatic aberration. This can be avoided by forming the grating on a curved (usually spherical) surface. The grating can then act as its own collimating and focusing components. For laboratory use spectroscope designs are often based upon the Rowland circle. This has a diameter equal to the radius of curvature of the grating, and the entrance slit, grating and focused spectrum all lie on the circle. For astronomical use such designs are usually too cumbersome, and more compact designs such as the Wadsworth (see later in this chapter) are preferred.

8.2.7 Echelle gratings

From equation (8.15) we may see that spectral resolution depends upon the order of the spectrum as well as the number of grating lines. An alternative to increasing the number of lines in order to improve resolution is therefore to observe the high order spectra. With a conventional grating, the high orders are at large angles to the normal to the grating, and shadowing by the vertical parts of the grating would obscure them completely. With an echelle grating, this is avoided and spectra with orders from several times ten to several times a hundred can be used. The echelle grating has the blaze angle greatly increased, and it is illuminated close to normal to the groove surfaces, and therefore at a large angle to the grating normal (figure 8.8). The coarser such a grating is made the higher the order it can operate at, for as may be seen from the diagram, the path difference between adjacent light beams is about twice the depth of the groove. For laboratory work, grooves may be a tenth of a millimetre deep, but for astronomical use somewhat finer grooves would be more normal.

Copyright © 1995 IOP Publishing Ltd.

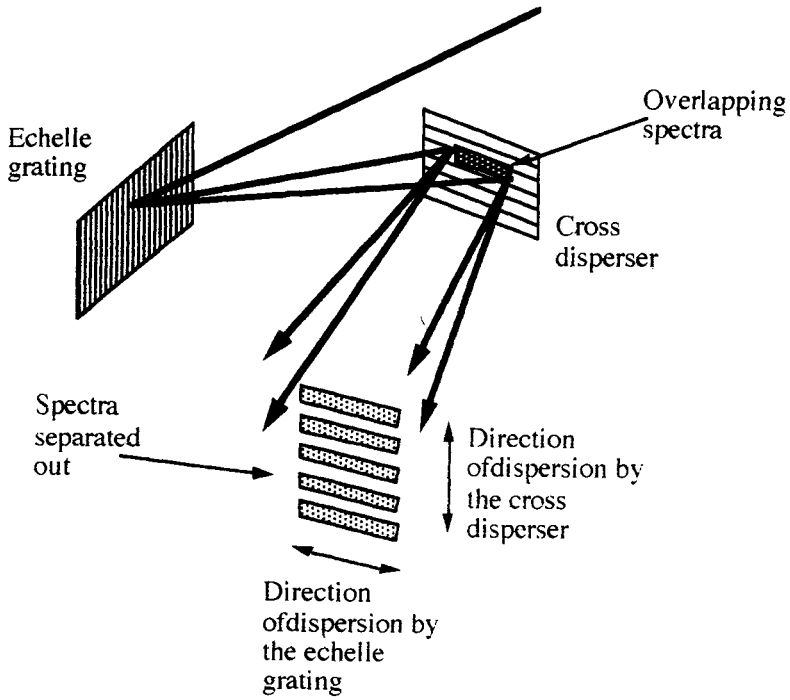

Figure 8.9 Schematic view of an echelle grating and a cross disperser.

The free spectral range (equation (8.16)) at high orders is very small, and the coarseness of the grating means that the individual apertures are quite large. The envelope of the diffraction pattern for a single aperture is therefore narrow. An individual order will thus cover only a small wavelength range before the modulation by the aperture diffraction pattern reduces its intensity to zero. It will then be overlapped by a different short wavelength interval from the next order spectrum. The final image thus consists of many short spectra superimposed onto each other. These are separated out by a second disperser acting perpendicularly to the direction of dispersion of the echelle (figure 8.9). This second disperser is called a Cross Disperser, and it can be of much lower dispersion than the echelle. The final spectrum then consists of an array of the short spectra (figure 8.10). Careful design of the system is needed to ensure that the short spectra overlap slightly in wavelength terms so that the complete spectrum can be synthesized.

8.3 GRATING PROBLEMS

We have already seen that the efficiency of a grating is reduced because it shares the available light amongst several orders, and that blazing can improve

Copyright © 1995 IOP Publishing Ltd.

that efficiency, but not back to 100%. A second problem is that of overlap of adjacent orders, and this has to be solved by filters if the free spectral range is large, or by a cross disperser if it is small. So far, however, we have assumed that the grating itself is perfect. That is to say that the individual rulings are all identical, equally spaced and parallel. In practice, any real grating will fall short of perfection, and its deficiencies will reduce the quality of the final spectrum.

The effects of grating deficiencies may most easily be envisaged via the spectroscope's instrumental profile, or point-spread function. The instrumental profile is the response of the instrument to an ideal monochromatic source. A perfect spectroscope would produce the spectrum of such a source as an emission line at a single wavelength. An optically perfect, but real spectroscope would have that emission line spread over a small range of wavelengths through diffraction effects at the various apertures within the instrument (figures 8.3 and 8.11), and this spread constitutes the instrumental profile. The width of the instrumental profile determines the spectral resolution (equation (8.15)), and so reduces as the number of grating lines increases and/or the spectral order increases.

The shape of the instrumental profile may be calculated from the Fourier Transform† of the grating. The perfect spectroscope can then be seen to require an infinite number of apertures, and the transmission or reflectance to vary sinusoidally across each aperture. The instrumental profile of the optically perfect but real spectroscope then has the shape of the Fourier transform for a regularly spaced finite set of delta functions, modulated by the Fourier transform for a single aperture with 100% transmission or reflectance across it (figure 8.3e).

It is easy to see how errors in the production of the grating will affect the instrumental profile, by considering the effect on the Fourier transform. The errors are of three main types. Random errors in the spacings and/or parallelism of the rulings will correspond to the function to be transformed containing

† Fourier Transforms are a mathematical means of splitting a function into its component frequencies. The Inverse Fourier Transform then recombines the frequencies back into the original function. One version of their mathematical expression is

Fourier transform:

$$F(s) = F(f(x)) = \int_{-\infty}^{\infty} f(x)\,\mathrm{e}^{-2\pi \mathrm{i} x s}\,\mathrm{d}x \tag{8.17}$$

Inverse Fourier transform:

$$f(x) = F^{-1}(F(s)) = \int_{-\infty}^{\infty} F(s)\,\mathrm{e}^{2\pi \mathrm{i} x s}\,\mathrm{d}s. \tag{8.18}$$

In the usual terminology, $f(x)$ is the function in the spatial domain, and its Fourier transform, $F(s)$, is the function in the frequency domain. The modulus of the Fourier transform is called the power spectrum, and is a plot of the relative contributions to the original function at each frequency. The shape of the power spectrum corresponds to the shape of the instrumental profile.

Copyright © 1995 IOP Publishing Ltd.

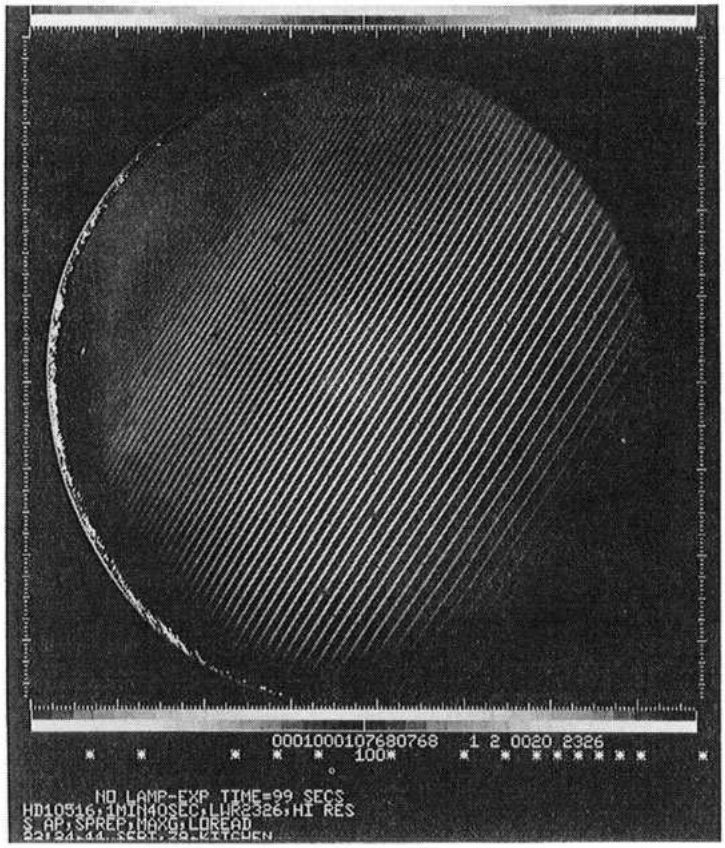

Figure 8.10 An echelle spectrum from the IUE spacecraft.

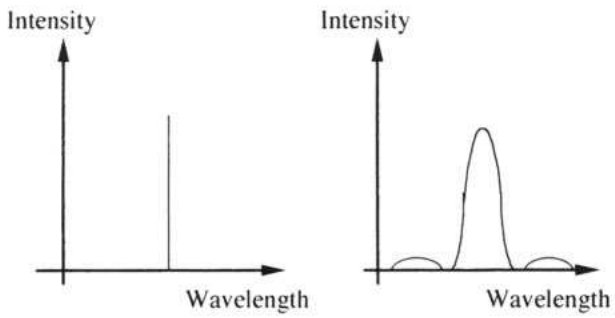

Figure 8.11 **Instrumental profiles** for a perfect spectroscope (left) and an optically perfect but real spectroscope (right).

Copyright © 1995 IOP Publishing Ltd.

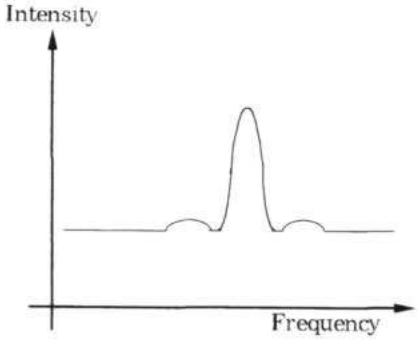

Figure 8.12 Power spectrum (i.e. instrumental profile) for a diffraction grating with random errors in the line spacing.

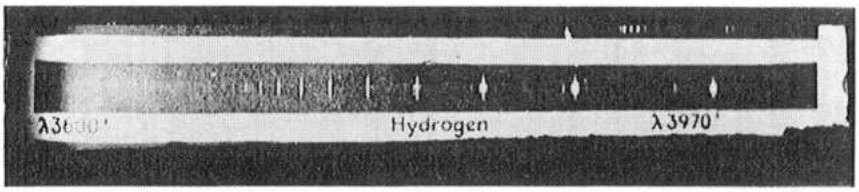

Figure 8.13 Spectrum showing Rowland ghosts due to a single periodic error in the grating rulings. Reproduced from *Fundamentals of Optics* (1976) by F Jenkins and H White with the permission of McGraw-Hill, Inc., New York.

white noise. The power spectrum of white noise is flat, and this will then have the transform for a regular grating superimposed (figure 8.12). Thus random errors will produce a background illumination, reducing the contrast in the final spectrum, but not affecting spectral resolution unless the errors become very gross. If the spacing and/or parallelism of the rulings vary in a periodic or multiply periodic manner, then the power spectrum will contain components corresponding to the frequencies and harmonics of those periodic errors. These will show up in the final spectrum as spurious spectral lines, known as Ghosts. A single periodic error produces Rowland ghost lines (figure 8.13) symmetrically about the true line and close to it.

Two or more incommensurate periodic errors produce Lyman ghosts which can occur at very different frequencies from the true line. Finally, there may be a progressive error in the rulings, so that their spacing increases or decreases in a regular manner. This has the effect of defocusing the collimated beam of light, so that it diverges or converges. It can largely be corrected by adjusting the focusing of the other optical components of the spectroscope.

Somewhat similar to ghosts, but not due to any deficiency of the grating, are the Wood's anomalies. These are sudden brightenings of the spectrum with a sharp onset and a slightly less sharp decline to longer wavelengths. The

Copyright © 1995 IOP Publishing Ltd.

anomalies arise from that energy that would go into spectral orders behind the grating, if these were possible, being redistributed back into the actually visible orders. The additional light is almost 100% plane-polarized perpendicular to the grating rulings, and so the existence of such anomalies can be checked through the use of a polaroid sheet.

8.3.1 Production of gratings

Gratings are originally made by drawing a diamond across an aluminium or magnesium blank. These metals are chosen because they are soft and so reduce the wear on the diamond. The shape of the grooves (figure 8.7) is obtained by shaping the point of the diamond. The machines that are used to rule gratings vary in their design, but typically the blank will be moved along one axis while the diamond moves at right angles to that motion. The position and rate of motion of both the blank and the diamond are monitored and controlled interferometrically. Positional accuracies of 3 nm and parallelism to 0.005 seconds of arc are achievable. Spectral resolution (equation (8.15)) requires gratings to have as many lines as possible. Wear of the diamond point, which leads to a change in the groove profile, limits the number of lines possible to a few times 10^5, depending upon the length of the grooves. The wear of the diamond is also affected by the loading that is placed on it. Thus many more lines can be ruled if the blaze angle is small than if it is large because the indentation required in the surface of the blank is reduced. Similarly, echelle gratings (above), requiring coarse deep grooves are limited in the number of grooves that can be produced.

Original gratings are rarely used directly. Instead, replicas of the original are made. A thin layer of plastic or resin is applied to the original, and stripped off after setting. The replica is then mounted onto a rigid substrate and aluminized. If a transmission grating is required then the replica is mounted onto a transparent substrate and not aluminized. Many replicas can be made from a single original, so reducing the cost of the grating.

A completely different approach to the production of gratings results in so-called Holographic Gratings. These have a sinusoidal form to the grooves, but are effectively unlimited in size and number of lines. They are produced by illuminating a layer of photoresist coated onto an optical flat with two lasers to produce straight line fringes. Etching the photoresist then leaves the grating on the optical flat.

8.4 DIFFRACTION GRATING SPECTROSCOPES

In order to produce a usable spectrum, the diffraction grating must usually be combined with several other components, and it is the combination that we refer to as a Spectroscope. Most designs of spectroscopes incorporate in one form

Copyright © 1995 IOP Publishing Ltd.

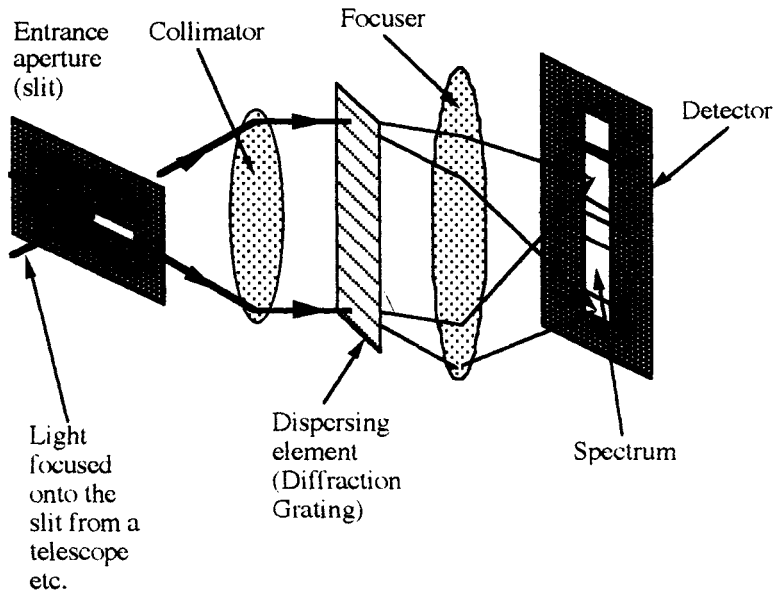

Figure 8.14 Basic components of a spectroscope.

or another the following basic components (figure 8.14): an Entrance Slit to reduce overlap between adjacent wavelengths and to reduce background noise, a Collimator to produce a parallel beam of light, a Disperser (in this case a diffraction grating), a Focusing Element to produce a focused image of the spectrum, and finally a Detector.

8.4.1 Resolution

A monochromatic source, as we have seen, will have an observed spectrum in which the light is spread to a greater or lesser extent into the neighbouring wavelength's positions. For a normal source with light of many wavelengths, the spectrum will therefore consist of an infinite number of such instrumental profiles, one for each wavelength present in the original spectrum, all overlapping each other. The spectral resolution (equation (8.15)) is set by the point at which the central peak of one instrumental profile is centred on the first dark fringe in the instrumental profile of a nearby wavelength. Such resolution can only be achieved in practice if it is not degraded by other components of the spectroscope. The collimator and focuser will normally be at least as large as the dispersing element. Their angular Rayleigh resolutions are given by

$$\alpha = \frac{1.22\lambda}{D} \qquad \text{radians} \tag{8.19}$$

Copyright © 1995 IOP Publishing Ltd.

where D is the diameter of the lens or mirror (assumed circular) in the same units as the wavelength (λ). The diffraction-limited resolutions of the collimator and focuser should not therefore degrade the spectral resolution of the disperser. However, their optical quality needs to be sufficiently adequate to ensure that their aberrations also do not significantly affect the image.

Generally of more significance for the spectral resolution is the width of the entrance slit. Let us consider the geometrical optics of the basic spectroscope (figure 8.14): since the beams for individual wavelengths exiting a good quality plane diffraction grating will still be collimated, the image of the slit will have a width, w, at the detector given by

$$w = \frac{f_f}{f_c} W \tag{8.20}$$

where W is the actual width of the slit, f_f is the focal length of the focuser and f_c is the focal length of the collimator. Clearly, the image of the slit must have a width no larger than that of the diffraction-limited instrumental profile for the remainder of the spectroscope, if it is not to degrade the spectral resolution. Thus from equations (8.14) and (8.15), we have

$$W < \frac{N f_c \lambda}{s \cos\theta} \tag{8.21}$$

as a constraint on the width of the entrance slit for it not to degrade the spectral resolution. For practical astronomical spectroscopes, this constraint often means that the slit width must be less than the size of a stellar image at the focus of the telescope. Therefore either some light must be wasted, or an image dissector (see below) used, or the slit width increased and some spectral resolution sacrificed.

A further limitation on spectral resolution is the size of the individual detecting elements (pixels) of the detector. These must be no larger than the physical size of the core of the instrumental profile when projected onto the detector, if resolution is to be preserved. For an optimum design of spectrometer, we should have all four constraints on spectral resolution equal to each other, i.e.

Spectral resolution of the disperser
= projected slit width
= optical resolution of all the components of the spectroscope
= detector pixel size.

Frequently astronomical spectroscopes have to take other constraints into account, such as that the physical size and weight of the spectroscope must be such that it can be mounted on the telescope, the focal ratio of the collimator must match that of the telescope, the focuser must be of short focal ratio to give reasonable exposure lengths, and by no means least, that the cost must remain within budget. All practical spectroscopes (chapter 10) therefore end up as compromises.

Copyright © 1995 IOP Publishing Ltd.

8.4.2 Throughput

Exposures to obtain the spectrum of an object with a given signal to noise ratio clearly need to be much longer than those to image the object directly to the same signal to noise ratio, because the available light is spread out into the spectrum. In practice, however, the required exposure is likely to be significantly longer than would be expected just on the basis of the greater area covered by the spectrum. This is because the spectroscope is not 100% efficient. If the entrance slit is smaller than the size of the stellar image at the telescope focus and an image dissector is not used, then the efficiency can fall to 10% or less. Spectroscopic exposures are then hundreds or thousands of times longer than those for direct images.

The efficiency of a spectroscope is measured by its Throughput, u (the terms Etendu and Light Gathering Power are also used). The throughput is defined as the amount of energy passing into the final spectrum when the entrance aperture is illuminated by unit intensity per unit area per unit solid angle. The throughput is given for spectroscopes with entrance slits by

$$u = A\tau\Omega \tag{8.22}$$

where A is the area of the aperture, τ is the fractional transmission of the optics of the spectroscope and Ω is the solid angle of the slit as seen from the collimator, i.e.

$$\Omega \approx A/f_c^2. \tag{8.23}$$

For slitless spectroscopes (see below), A is the area of the collimator, or focuser, or effective area of the dispersing element, whichever is the smaller, while Ω is the solid angle actually accepted by the spectroscope–telescope combination.

Unless only a very small spectral range is required, any transmission optics cannot have anti-reflection blooming applied because this is wavelength-dependent. There will thus be a light loss of at least 5% at each transmission surface. Mirror surfaces can be re-aluminized regularly, and so losses kept to 5% to 10% at each reflection. The widely used reflection grating, however, cannot be re-aluminized, and so losses due just to reflection efficiency could be from 10% to 20%. Shadowing, and energy going into unwanted orders could add another 10% to 50% to these losses. Thus for a Wadsworth spectroscope (see below) with an efficient grating, we might expect τ to be up to 0.7. But for a spectroscope using the not uncommon design of an achromat for the collimator, and a Schmidt camera for the focuser, with an older or less efficient grating, τ could fall to 0.3 or less. Losses due to the stellar image being larger than the slit are additional to these due to the optics and are not included in the figure for the throughput.

For a typical spectroscope carried at the Cassegrain focus of a major telescope, we might have a slit one or two hundred microns wide, and a few millimetres long, and f_c from 0.1 m to 0.5 m. The throughput would thus be

Copyright © 1995 IOP Publishing Ltd.

in the range 10^{-9} to 10^{-14} for SI units. For a spectroscope at the Coudé or Nasmyth focus of a telescope, the collimator could have a focal length up to 10 m, resulting in throughputs down to 10^{-16}.

A useful parameter for comparing the overall performances of spectroscopes is the product of spectral resolution and throughput

$$P = Ru. \tag{8.24}$$

Since spectroscopes for use at the (fixed) Coudé focus usually have much greater spectral resolutions than those at the (moving) Cassegrain focus of a telescope, their overall performances become comparable. Other things being equal, P will generally be highest for Fabry–Pérot etalon-based spectroscopes (see below), and lowest for prism-based spectroscopes, with diffraction gratings coming between.

8.4.2.1 Limiting magnitude. The faintest object from which a useful spectrum may be obtained determines the limiting magnitude of the spectroscope–telescope combination. Its value will clearly depend upon the throughput, but also upon many other factors. Thus the detector to be used makes a considerable difference, with CCDs having upwards of a hundred times the quantum efficiency of photographic emulsions. With most detectors, however, sensitivity varies with wavelength, so that basic CCDs and other silicon-based detectors become only a few times better than photographic emulsions in the blue part of the spectrum. With some detecting systems, such as the IPCS (Image Photon Counting System), noise reduction techniques permit given signal to noise ratios to be reached more rapidly than with other detectors of equal quantum efficiency. Secondly it will depend upon the purpose for which the spectrum is required, and the signal to noise ratio needed. Thirdly it will depend upon the nature of the spectrum; a nebula can have its spectrum found with surprising ease in comparison with that for a star since the light from the former is concentrated into a few strong emission lines with little or no continuum emission. The commonly used integrated magnitude for extended objects such as nebulae and galaxies will, by contrast, be far too optimistic since only a small fraction of the object will be covered by the spectroscope's entrance slit.

Finally, but by no means least, limiting magnitude will depend upon the patience of the observer and the generosity of the panel awarding the telescope time! Spectroscopic exposures generally have a low background and the observer can therefore continue to integrate the signal for long periods. Hubble, when he obtained the spectra of galaxies, used exposures extending over several nights. While such long exposures are unlikely today, since CCD exposures are limited by cosmic ray noise, the equivalent exposure can be built up by summing several spectra from shorter exposures after the cosmic ray spikes have been removed.

Clearly, no precise value for the limiting spectroscopic magnitude is possible. However, as a guide for comparing different spectroscope–telescope

Copyright © 1995 IOP Publishing Ltd.

combinations, which is useful when the star's image is larger than the slit and it is trailed along the slit to widen the spectrum, we have Bowen's formula:

$$m_{\text{limit}} = 12 + 2.5 \log \left(\frac{W D_c D_t g Q t (\mathrm{d}\lambda/\mathrm{d}\theta)}{f_c f_f \alpha H} \right) \tag{8.25}$$

where m_{limit} is the faintest B magnitude giving a usable spectrum, D_c is the diameter of the collimator and D_t is the diameter of the telescope's objective. When the slit is wider than the star's image, the exposure will vary as D_t^2. g is the optical efficiency of the whole spectroscope–telescope combination. It therefore includes τ, and typically has values in the range 0.1 to 0.3. Q is the fractional quantum efficiency of the detector, t is the exposure time in seconds, α is the angular size of the star's image, typically 5 to 20 μrad ($1''$ to $4''$), and H is the height of the spectrum.

For extended objects the situation is much more complicated, and different limiting magnitudes will be found for different types of object. The exposures required will generally need to be found by experiment. For a given object, however, it is useful to remember that the exposure for a given signal to noise ratio will vary as the square of the focal ratio of the telescope.

8.4.3 Designs of diffraction grating spectroscopes

The basic spectroscope shown in figure 8.14 can be used as illustrated, or used with a reflection grating in place of the transmission grating and/or with one or both of the collimator and focuser replaced by reflection optics. Reflection optics and grating are generally to be preferred over a transmission grating and refractive optics because of the intrinsic achromatism of reflection. Most astronomical spectroscopes employ plane diffraction gratings in either the basic design or Littrow (figure 8.15) or Ebert (figure 8.16) designs.

Saving on components and therefore also on light losses and cost can be made by the use of a curved reflection grating. As previously mentioned, designs based upon the Rowland circle are used for laboratory spectroscopes (e.g. figure 8.17). Normally, however, these designs are too cumbersome to be suitable for astronomical use. Not only are they physically large and heavy and therefore difficult to mount on a telescope, but their size makes it impossible to avoid differential flexure as the gravitational loading on the instrument changes with the telescope's attitude. The precise alignment required for the components of the spectroscope will thus be ruined. A commonly encountered design is therefore the Wadsworth (figure 8.18). At the price of an extra reflection, this gives a much more compact instrument.

Echelle gratings can be incorporated into designs similar to those above, but with the additional need, normally, for a cross disperser. One neat design, based upon the Littrow, simply places a prism immediately in front of the echelle grating (figure 8.19).

Copyright © 1995 IOP Publishing Ltd.

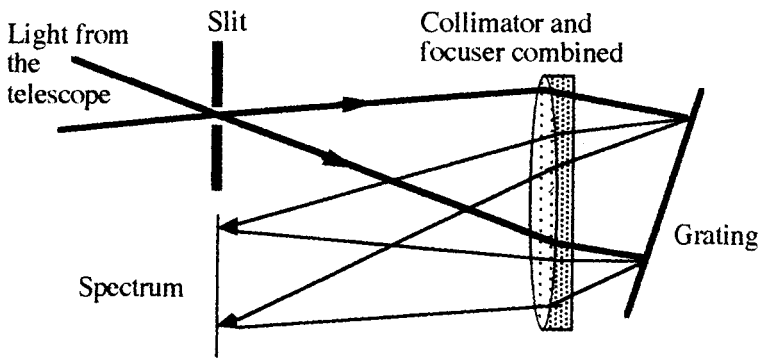

Figure 8.15 The Littrow design for a grating spectroscope.

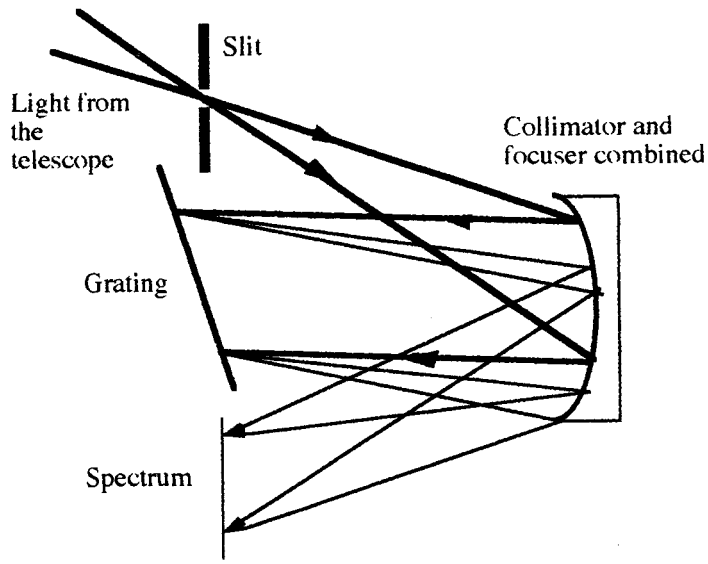

Figure 8.16 The Ebert design for a grating spectroscope.

For extended sources, a much longer slit than normal may be used for the spectroscope. The spectrum is then not trailed and so a particular horizontal section of the spectrum corresponds to a particular part of the image (figure 8.20). The spectra of many points across the image can thus be acquired in one exposure. Long slit spectroscopes do not differ in principle from the designs discussed above, but more consideration needs to be given to the performance of the instrument away from its optical axis. In particular, astigmatism and curvature of the spectral lines may become important.

All designs so far considered employ a slit as the entrance aperture to the

Copyright © 1995 IOP Publishing Ltd.

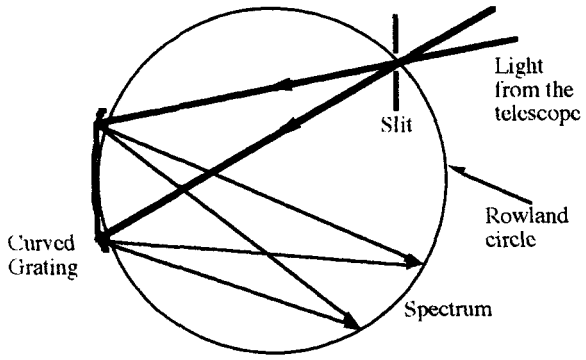

Figure 8.17 Design for a grating spectroscope based upon a curved diffraction grating and the Rowland circle.

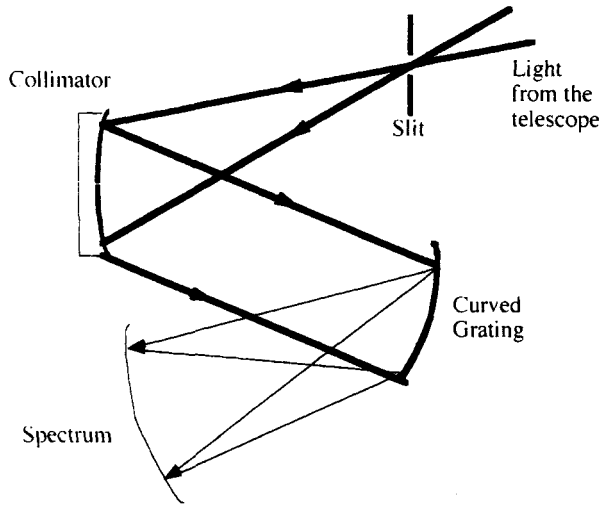

Figure 8.18 The Wadsworth design for a grating spectroscope.

spectroscope. While this has many advantages, it can also result in significant light losses. For some purposes, the slit can be dispensed with, resulting in a much more efficient system. Slitless spectroscopes come in two classes. The first has the disperser placed before the telescope objective. Normally a prism is used for this purpose since all the light is then concentrated into one spectrum. Such objective prism spectroscopes are considered later in this chapter. A transmission grating could be used in a similar fashion, but since it would be almost impossible to blaze, it would be very inefficient. Very coarse objective gratings are, however, sometimes employed in astrometry, since it can be easier to measure the positions of the first-order spectra which appear as faint star-

Copyright © 1995 IOP Publishing Ltd.

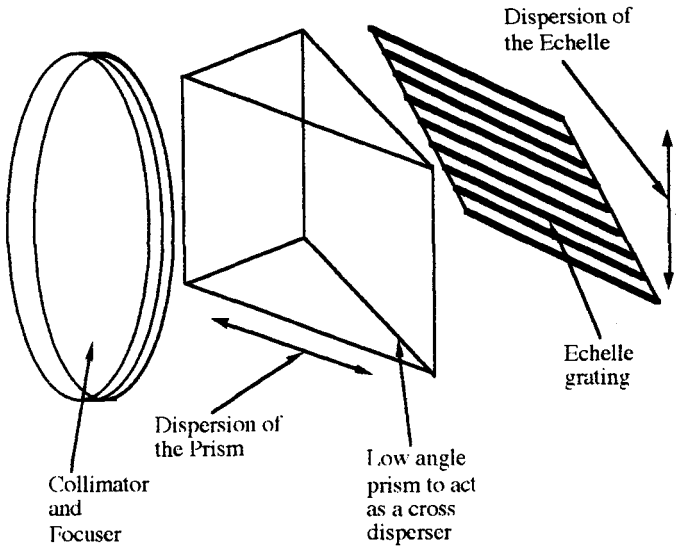

Figure 8.19 One design for a cross disperser for use with an echelle diffraction grating in a Littrow-type spectroscope.

like images on either side of the zero-order image than that of the overexposed zero-order image.

The other class of slitless spectroscopes contains standard spectroscope designs without the entrance slit, or with a slit larger than the size of the image at the focal plane of the telescope. For stellar images, this would normally degrade the final spectrum into near uselessness. It is therefore something of a paradox that such spectroscopes are actually used for extended objects. Slitless spectroscopes can be used when the spectrum of the object has little or no continuum, but is simply composed of emission lines. Such spectra occur for many objects including planetary nebulae, H II regions, supernova remnants, the solar chromosphere etc (chapters 14 and 16 and figure 1.4). The final spectrum then consists of a series of monochromatic images of the object in the light of each of its emission lines.

Grating spectroscopes operating outside the visual region (370–700 nm), but still within the optical region (100 nm to 20 μm) do not differ much in their optical principles from the standard designs discussed so far. Their differences lie in matching the details of the design to the requirements for the new wavelengths. Thus clearly the detector to be used must be able to respond to the wavelengths forming the spectrum. In the ultraviolet, the constraints on the optical components are more stringent because of the $\lambda/8$ requirement on mirror surface accuracies if the Rayleigh resolution is not to be degraded. The reflectivity of aluminium decreases with wavelength, and is down to 70% at 200 nm, so other surface coatings may need to be used. If transmission

Copyright © 1995 IOP Publishing Ltd.

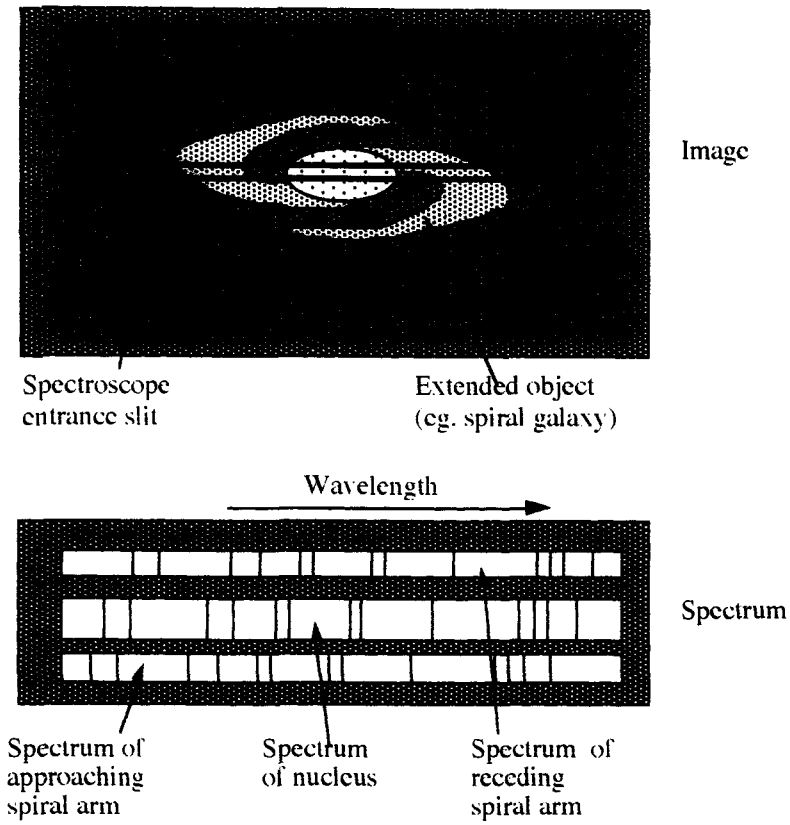

Figure 8.20 Long slit spectroscopy.

optics are used, then they must be fabricated from materials such as fused quartz or rock salt because glass absorbs at short wavelengths. Not least, for an instrument operating at wavelengths shorter than about 350 nm, it must be flown on a rocket, spacecraft, balloon or high altitude aircraft to get it above some or all of the Earth's atmosphere, which would otherwise absorb the radiation. In the infrared, similar considerations apply. The increased wavelength, however, reduces the requirements on surface accuracy for the optical components. At the longer infrared wavelengths, it may be necessary to cool the grating or even the whole instrument in order to reduce noise levels. Materials such as germanium, fluorite or proustite need to be used for transmission components. Although the atmosphere does not absorb completely over the infrared region, it will still be highly advantageous to use the infrared spectroscope from a high altitude observatory or to lift the instrument even higher by balloon etc.

Examples of specific designs for spectroscopes in current use are discussed in chapter 10.

Copyright © 1995 IOP Publishing Ltd.

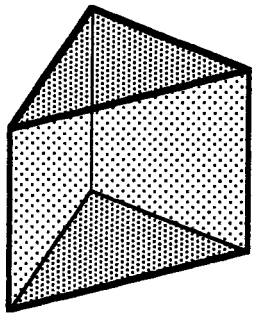

Figure 8.21 A prism as used in spectroscopy.

8.5 PRISMS

The classical and earliest spectroscope design employs a prism (chapter 1), and is the type most likely to have been encountered at school etc. Strictly the term 'prism' denotes an object with a constant cross section in one axis. In spectroscopic and common usage, however, it refers only to such an object with a triangular cross section (figure 8.21). It would be unusual to design an astronomical spectroscope today which used a prism as the primary disperser. This is because the prism needs to be large to have a reasonable spectral resolution. A large prism would be expensive, heavy, and would absorb the radiation passing through it. Moreover, comparable spectral resolution could be achieved with a quite cheap diffraction grating. Older spectroscopes which do employ prisms are still in use, and a prism may be used as a cross disperser (see echelle gratings above and Fabry–Pérot etalons below) or as an objective prism (see below). A brief discussion of their properties is therefore included here.

A prism acts as a disperser through the effect of differential refraction. That is to say, the refractive index of the material from which it is composed varies with wavelength, and so rays of differing wavelengths are deviated to different extents on passage through the prism (figure 8.22). From the law of refraction (Snell's law):

$$\frac{\sin i}{\sin r} = \mu \tag{8.26}$$

where μ is the refractive index, i is the angle of incidence of the ray to the normal to the surface and r is the angle of refraction of the ray to the normal to the surface, we may easily derive the equation for the deviation of a ray of particular wavelength (and therefore for a particular value of μ):

$$\theta = i - \alpha + \sin^{-1}\left\{\mu \sin\left[\alpha - \sin^{-1}(\sin i/\mu)\right]\right\} \tag{8.27}$$

where θ is the deviation (figure 8.22) and i is the angle of incidence to the normal to the first surface of the prism.

Copyright © 1995 IOP Publishing Ltd.

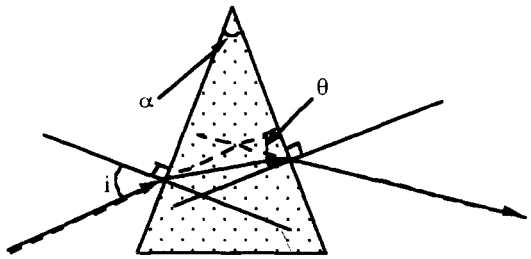

Figure 8.22 Deviation of a light ray by a prism.

Table 8.1 Refractive indices of some optical materials.

	Wavelength (μm)						
Substance	0.2	0.4	0.6	0.8	1.0	1.5	2.0
Crown glass		1.530	1.516	1.510	1.506	1.500	
Dense flint glass		1.652	1.618	1.608	1.603		
Fused quartz	1.600	1.470	1.457	1.452			
Fluorite	1.496	1.442	1.431	1.430	1.429	1.427	1.425
Rock salt	1.890	1.570	1.544	1.537	1.532	1.528	1.525

For the often utilized situation with the ray passing through the prism symmetrically (i.e. parallel to the base—the condition for minimum deviation and highest image quality), the equation simplifies to

$$\theta = 2\sin^{-1}\left[\mu \sin(\alpha/2)\right] - \alpha \tag{8.28}$$

and simplifies even further for the normally used apex angle of 60° to

$$\theta = 2\sin^{-1}(\mu/2) - 60^\circ. \tag{8.29}$$

Typical values for the refractive indices of some relevant optical materials are listed in table 8.1.

Inspection of equation (8.28) and table 8.1 shows that, since the change in refractive index with wavelength decreases with increasingly wavelength for all the optical materials, a prism-based spectroscope will become increasingly ineffective at longer wavelengths. Thus even though rock salt and fluorite transmit in the infrared, they would not generally form useful dispersers for infrared spectroscopes.

The change in refractive index with wavelength may be described by the empirical Hartmann formula:

$$\mu_\lambda \approx A + \frac{B}{\lambda - C} \tag{8.30}$$

Copyright © 1995 IOP Publishing Ltd.

Table 8.2 Hartmann constants.

	A	B	C
Crown glass	1.500	3.5×10^{-8}	-2.5×10^{-7}
Dense flint glass	1.650	2.1×10^{-8}	1.5×10^{-7}
Fluorite	1.429	5.3×10^{-10}	3.6×10^{-7}

where A, B and C are the Hartmann constants for a particular material. Typical values are listed in table 8.2.

The dispersion of a 60° prism is then given for glass-like materials by

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} \approx \frac{-57.3AB}{(\lambda - C)^2} \quad {}^\circ\,\mathrm{m}^{-1}. \tag{8.31}$$

The spectral resolution is determined by the spread function due to diffraction at the apertures represented by the entry and exit faces of the prism. For symmetrical passage of the light through the prism, these apertures are equal and produce a rectangular light beam with a width of

$$D = L\left[1 - \mu^2 \sin^2(\alpha/2)\right]^{1/2} \tag{8.32}$$

where L is the length of the face of the prism. The Rayleigh resolution for such a rectangular beam is λ/D, and so we find that the spectral resolution is given by

$$R \approx \frac{ABL\left[1 - 0.25\left(A + \frac{B}{\lambda - C}\right)^2\right]^{1/2}}{(\lambda - C)^2}. \tag{8.33}$$

For a reasonably typical prism for use in an astronomical spectrograph made of dense flint and with a side length of 0.1 m, this therefore gives a spectral resolution of around 15 000, or less than half that of an equivalent diffraction grating.

8.6 PRISM SPECTROSCOPES

A slit-type prism spectroscope design can be produced for most of the diffraction grating instruments discussed earlier by replacing the grating with the prism. The most commonly encountered such designs are the basic spectroscope (figure 8.14) and the Littrow spectroscope (figure 8.15). For the Littrow design, a 30° prism would be used, with the rear surface aluminized. The light is therefore reflected and passes twice through the prism making it equivalent to a single 60° prism. The reflected beam is returned nearly along the same direction as the incoming beam, and so passes through the system in a similar manner to

Copyright © 1995 IOP Publishing Ltd.

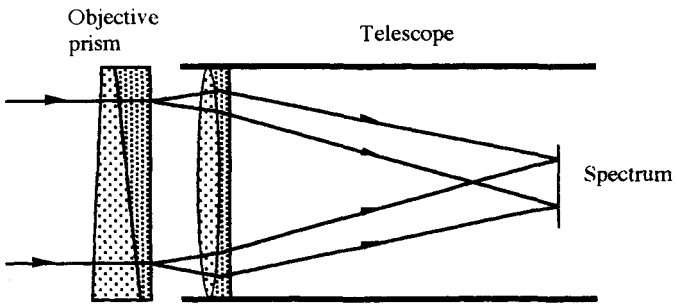

Figure 8.23 An objective prism spectroscope.

that shown in figure 8.15. There is no equivalent to the curved gratings, so the Rowland circle and Wadsworth designs have no prism-based equivalents.

A prism can have a diffraction grating applied to one of its surfaces, and the combination is then known as a Grism. In one form it provides a direct vision spectroscope for solar work, with the prism's deviation being equal and opposite to that of the blazed order of the transmission grating on its surface. The light then has no net deviation and is only dispersed, which simplifies the design of the whole instrument by eliminating off-axis components.

The simplest spectroscope of all is the objective prism. In this design, a thin prism covers the whole of the objective of the telescope, and so every object in an image is replaced by its spectrum. A simple prism may be used, in which case the telescope has to be pointed at a large angle to the desired direction because of the deviation caused by the prism. Alternatively two prisms of different glasses may be used in opposition to give zero deviation while still retaining some dispersion (figure 8.23). This is the exact inverse of the achromatic lens, where lenses of two different glasses are combined to give (close to) zero dispersion while still retaining focusing (i.e. deviation). The objective prism is a very efficient design because not only does it dispense with the slit and so improve the throughput hugely (see earlier discussion), but many spectra can be obtained in a single exposure. If an objective prism is applied to a Schmidt camera, then upwards of 10^5 spectra can be obtained on a single plate.

Unfortunately, the objective prism spectroscope also has serious disadvantages. The most important is the difficulty of finding radial velocities from the spectra because there is no comparison spectrum (chapter 9). Various attempts to solve this difficulty have been made, of which the most successful is to take two exposures on the same plate, with a slight displacement of the telescope and with the prism reversed between the exposures. Each object in the image then has two adjacent spectra (figure 8.24). Radial motions of the objects will give rise to Doppler shifts, causing the separation of a particular line in one spectrum from the same line in the second spectrum to change. If some of the objects have radial velocities known already, then that change in separation can be calibrated to give the velocities of all the objects on the plate. Other

Copyright © 1995 IOP Publishing Ltd.

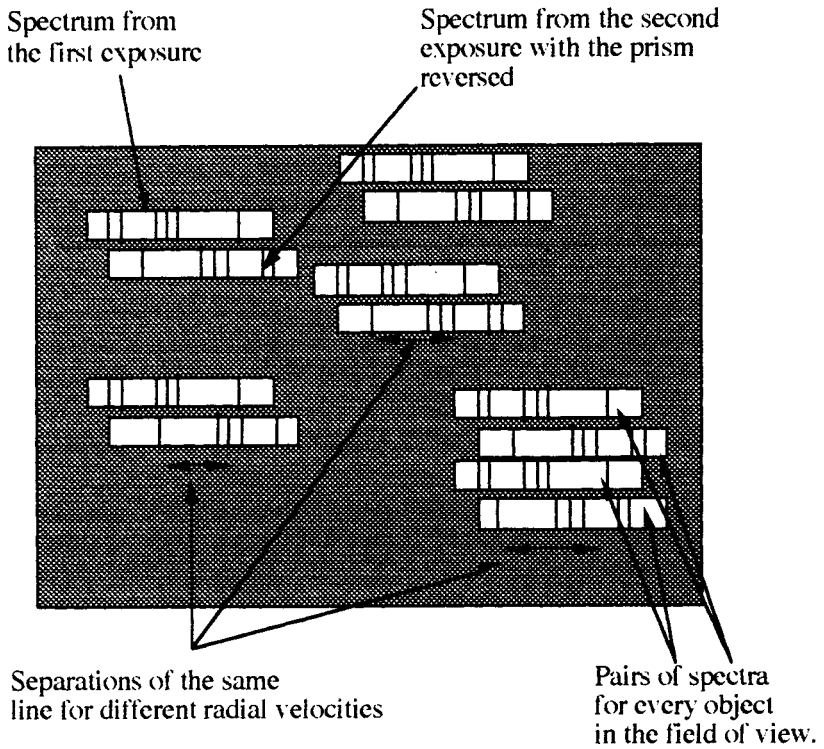

Figure 8.24 Doubly exposed objective prism plate showing the change in the separation of identical lines in pairs of spectra with the changing radial velocities of the objects.

drawbacks include the low dispersion of the spectra, and the weight and cost of the prism for a large telescope.

8.7 FOURIER TRANSFORM SPECTROSCOPE (MICHELSON INTERFEROMETER)

The interferometer used by Michelson and Morley (figure 8.25) in their classic experiment of 1887 to try to detect the Earth's motion through the aether can form the basis of a type of spectroscope known as a Fourier Transform Spectroscope. Unlike the diffraction grating or prism, the spectrum is not produced directly. The operating principle, however, can be understood from a couple of examples.

Light from a monochromatic source will clearly form a simple interference pattern at the focus of the instrument. If the detector accepts only a narrow portion of that pattern, then its output will vary as the position of the moving mirror changes and different parts of the pattern impinge onto it. If the moving

Copyright © 1995 IOP Publishing Ltd.

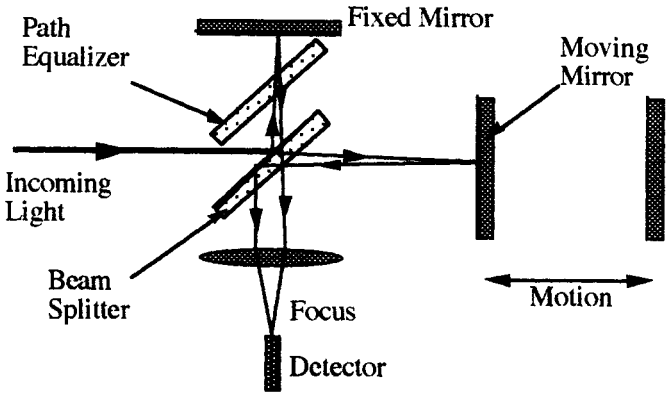

Figure 8.25 Optical components of a Fourier transform spectroscope.

mirror is scanned smoothly, then the detector output will be a simple sine wave (figure 8.26). Monochromatic light of a slightly different wavelength will give a similar output with a slightly different period. A single source in which both wavelengths are present will have these two outputs combined by straightforward addition, to give an output with a beat frequency (figure 8.27).

In frequency terms, a monochromatic source is a single delta function, and its Fourier transform (equation (8.17)) is a sine wave. A double delta function, corresponding to the bichromatic source, would have a Fourier transform with a beat frequency. We may see therefore that for more complex spectra, the output from the instrument will be related to the Fourier transform of the spectrum. Conversely, the spectrum may be found from the output through the inverse Fourier transform (equation (8.18)). Hence the name given to the instrument. The precise relationship between spectrum and output is obtained from the real part of the inverse transform:

$$I(\lambda) \propto \int_0^\infty I(\Delta P) \cos\left(\frac{2\pi \Delta P}{\lambda}\right) \mathrm{d}\,\Delta P \tag{8.34}$$

where $I(\lambda)$ is the intensity in the spectrum at wavelength, λ, ΔP is the path difference between the two beams for a particular position of the moving mirror and $I(\Delta P)$ is the output (intensity) from the instrument for a particular value of ΔP.

Since there is no spectrum produced directly, there is no equivalent to the dispersion of a diffraction grating or prism. The spectral resolution of the Fourier transform spectroscope is theoretically infinite. In practice, however, the path difference cannot be extended to infinity as required by equation (8.34), and this, together with the measurements having to be made discretely instead of continuously, imposes a limit on the spectral resolution. The spectral resolution is thus given by

$$R = 2\Delta P_{\max}/\lambda \tag{8.35}$$

Copyright © 1995 IOP Publishing Ltd.

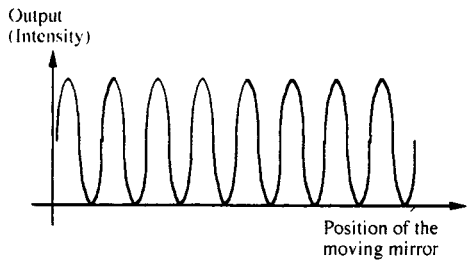

Figure 8.26 Output from a Fourier transform spectroscope for a monochromatic source.

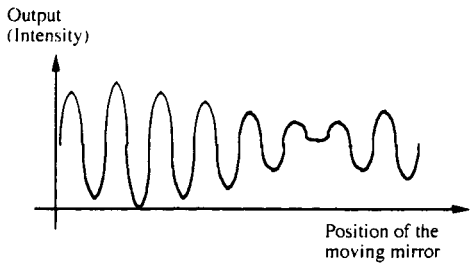

Figure 8.27 Output from a Fourier transform spectroscope for a bichromatic source.

where ΔP_{max} is the maximum path difference for the two beams that the instrument can achieve. For laboratory spectrometers, ΔP_{max} can be as much as two metres, giving a spectral resolution in the near infrared, where they are usually used, of over a million. For astronomical use, Fourier transform spectroscopes would normally be of smaller size and lower resolution than this, but some instruments used for solar spectroscopy do have resolutions of up to a hundred thousand.

The limit to the spectral resolution of a Fourier transform spectroscope means that there is no need to make continuous measurements of the output. This in any case is impossible, since even apparently continuous devices, like chart recorders, are in effect digital with sampling intervals given by their time constants. The sampling interval, and hence total number of measurements, for the Fourier transform spectroscope should be adequate to retain the spectral resolution, but not more than this or time and money will be being wasted. The sampling interval thus required is given by

$$\Delta x = \frac{(\lambda_1 + \lambda_2)^2}{16(\lambda_1 - \lambda_2)} \tag{8.36}$$

where Δx is the distance moved by the moving mirror between successive measurements of the output, and λ_1 and λ_2 are the short and long wavelength limits to the spectrum. Thus a Fourier transform spectroscope to obtain a spectrum from 1.0 to 1.2 μm at a spectral resolution of 10 000, would require

Copyright © 1995 IOP Publishing Ltd.

a maximum path difference of 5.5 mm for the two light beams, and therefore a movement of the mirror by 2.75 mm. A total of 1800 measurements would be needed at 1.5 μm intervals for the moving mirror.

The Fourier transform spectroscope has two big advantages over most other types of spectroscope. These are known as the Multiplex (or Fellgett) and Throughput (or Jacquinot) advantages. The multiplex advantage occurs when spectroscopy in a particular wavelength region is limited by noise in the detector. As the name implies, this advantage of the Fourier transform spectroscope arises because the detector is viewing the whole spectrum all the time. The average intensity detected will be about half the integrated intensity of the whole spectrum. If a spectrum with N resolution elements is to be obtained, then the total time taken to obtain the spectrum if a grating or prism spectrum is scanned by the detector would be

$$T_{\text{scan}} = Nt \tag{8.37}$$

where t is the time taken to obtain a single measurement to the required signal to noise ratio. With a Fourier transform spectroscope, the same spectrum would require twice as many measurements (because the inverse transform gives both $I(\lambda)$ and $I(-\lambda)$), but the increased intensity being viewed by the detector would reduce the time needed to reach the required signal to noise ratio. The time for each measurement would thus be

$$\frac{t}{\sqrt{N/2}} \tag{8.38}$$

and the total time required to obtain the same spectrum using the Fourier transform spectroscope would be

$$T_{\text{FTS}} = \sqrt{8N}\, t. \tag{8.39}$$

The multiplex advantage is thus

$$T_{\text{scan}}/T_{\text{FTS}} = \sqrt{N/8}. \tag{8.40}$$

For the previous example of a spectrum from 1.0 to 1.2 μm, the Fourier transform spectroscope would obtain it in one fifteenth of the observing time required for a scanning instrument.

The multiplex advantage of the Fourier transform spectroscope disappears when the system is not limited by detector noise or if an array of detectors is used. The throughput advantage, however, remains. This advantage is the result of the larger entrance aperture allowed by the interferometer since no aperture is required to preserve spectral resolution; for a given resolution, the light grasp can be some two orders of magnitude larger than that of a grating or prism spectrometer.

There is a further advantage to the Fourier transform spectroscope which arises as a consequence of what at first sight seems to be one of its disadvantages.

Copyright © 1995 IOP Publishing Ltd.

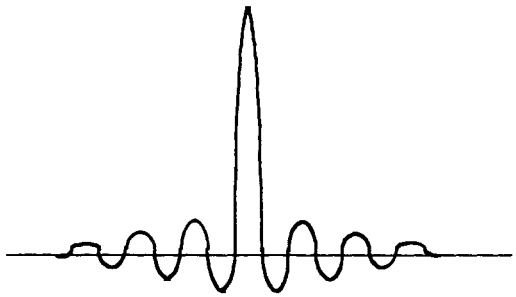

Figure 8.28 Natural instrumental profile of a Fourier transform spectroscope.

The Fourier transform spectroscope does not give a spectrum directly, but this has to be computed after the observations by using the inverse Fourier transform. Even with the fast Fourier transform algorithm, the delay between observation and spectrum can be significant. If observing from a remote site with inadequate computing facilities, the delay could be for days or more until the observer returns to base, thus eliminating the possibility of chasing up interesting details from the 'first look' at the data. The advantage that comes from the requirement of processing the output in order to obtain the spectrum is that it allows some image processing to be included in the data reduction. Of greatest significance in this respect is the ability to optimize the instrumental profile (or point-spread function) of the instrument. The natural instrumental profile of the Fourier transform spectroscope is of the form

$$\frac{\sin \Delta\lambda}{\Delta\lambda} \tag{8.41}$$

where $\Delta\lambda$ is the distance from the central wavelength. This instrumental profile has fringes (figure 8.28), which can lead to spurious features appearing near very strong emission or absorption lines or edges. By weighting the data during the inverse transformation, the form of the instrumental profile can be changed, though usually at the cost of some slight reduction in the Rayleigh resolution. The commonest weighting function in use is triangular:

$$\omega(\Delta P) = 1 - \frac{\Delta P}{2\Delta x} \tag{8.42}$$

which reduces the secondary maxima to those of a rectangular aperture, while the Rayleigh resolution deteriorates by a factor of two. Such a purposeful manipulation of the instrumental profile is part of the technique called Apodization.

8.8 FABRY-PÉROT ETALONS

The last form of disperser of current interest to the astronomer is the Fabry–Pérot Etalon. This is a classic disperser in the manner of the diffraction

Copyright © 1995 IOP Publishing Ltd.

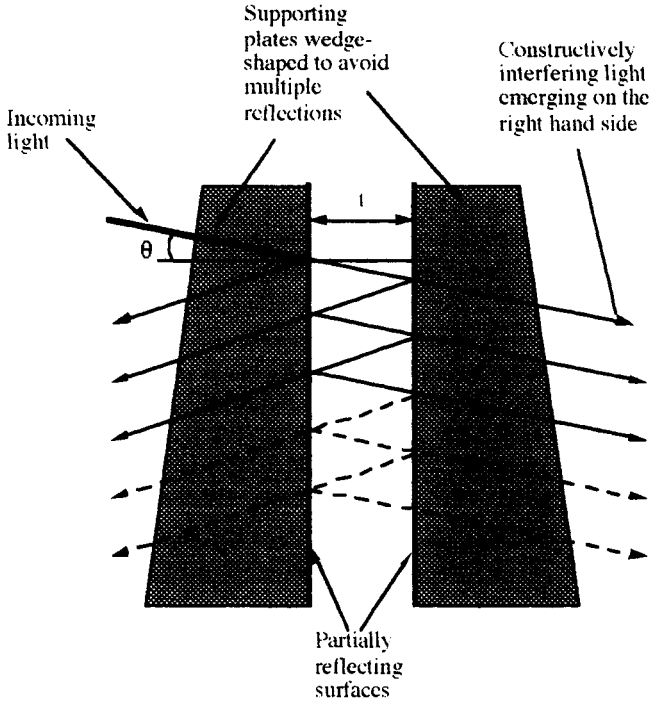

Figure 8.29 The optical paths in a Fabry–Pérot etalon.

grating or prism, producing a spectrum directly, and can be incorporated in their place into 'normal" spectroscope designs. It consists of two partially reflecting parallel surfaces (figure 8.29). Rays emerging from the right with integer number of wavelength path differences interfere constructively. Other wavelengths experience destructive interference, and so do not emerge on the right. Since the very first reflection is an internal one, and so has a 180° phase difference from the external reflections, the rays emerging on the left have a total destructive interference when there is constructive interference on the right hand side, and vice versa.

The path difference between successive rays is given by (figure 8.29)

$$\Delta P = 2t\cos\theta. \tag{8.43}$$

The beam of light emerging on the right hand side will thus only contain those wavelengths for which ΔP is an integer number of wavelengths. The number of interfering beams, and hence the purity of the emerging wavelengths, depends upon the number of reflections. If the reflectivity of the partially reflecting surfaces is high then there will be many reflections and the purity of the emerging beams will be good (figure 8.30). Typically a reflectivity of 90% or so is used

Copyright © 1995 IOP Publishing Ltd.

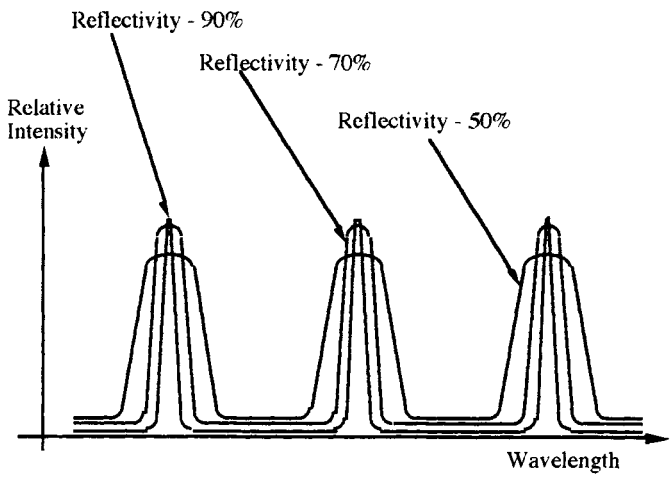

Figure 8.30 The output with respect to wavelength from a Fabry–Pérot etalon for a point source of white light.

in practical devices. The principal maxima shown in figure 8.30 will be at wavelengths given by

$$\lambda = \frac{2t\cos\theta}{m-1}, \frac{2t\cos\theta}{m}, \frac{2t\cos\theta}{m+1}, \ldots \tag{8.44}$$

where m is an integer (the order of the spectrum). However, in the case of a point source, they will not be physically separated from each other, unless a cross disperser is used.

Using an entrance slit as the source, the angle of incidence onto the etalon will vary with position along the slit. For a monochromatic source, therefore, there will be a number of maxima accordingly as

$$\theta = \cos^{-1}\left(\frac{(m-1)\lambda}{2t}\right), \cos^{-1}\left(\frac{m\lambda}{2t}\right), \cos^{-1}\left(\frac{(m+1)\lambda}{2t}\right), \ldots \tag{8.45}$$

corresponding to constructive interference at the one wavelength in different order spectra (figure 8.31), and these will be physically separated in the image.

A second monochromatic slit source with a different wavelength from the first would have a similar output to figure 8.31, with the physical positions of the maxima altered slightly. A bichromatic slit source would therefore have an output like that shown in figure 8.32. Finally a white light slit source would thus have an output from the etalon comprising a series of short spectra of differing orders superimposed onto each other. A cross disperser can then convert these into an array, and careful design of the system's parameters can lead to coverage of the complete spectrum, as for the echelle grating (above). The free spectral range of an etalon is given by equation (8.16).

Copyright © 1995 IOP Publishing Ltd.

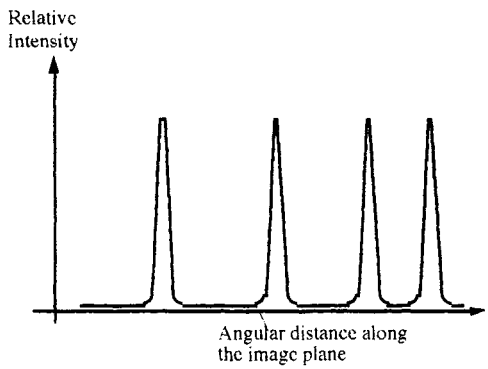

Figure 8.31 Output from a Fabry–Pérot etalon for a monochromatic slit source.

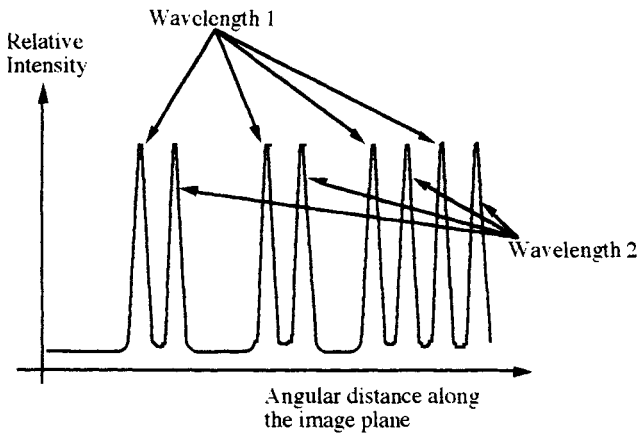

Figure 8.32 Output from a Fabry–Pérot etalon for a bichromatic slit source.

The dispersion of the etalon may be found by differentiating equation (8.44):

$$\frac{d\lambda}{d\theta} = \frac{2t\mu \sin\theta}{m} \tag{8.46}$$

where μ is the refractive index of the material between the plates of the etalon. From equation (8.44), and for the usual case of θ small and air between the plates, so that $\mu \approx 1$, we get

$$\frac{d\lambda}{d\theta} \approx \lambda\theta. \tag{8.47}$$

The equivalent of the Rayleigh spectral resolution for an etalon may be shown to be given by

$$R \approx \frac{2\pi t\sqrt{r}}{(1-r)\lambda} \tag{8.48}$$

Copyright © 1995 IOP Publishing Ltd.

where r is the fractional reflectivity of the surfaces of the etalon. In the laboratory, etalons may have spacings of several tens of centimetres and resolutions of several million. More modest instruments are in astronomical use, typically with separations of a few hundred microns and resolutions of a few times ten thousand or less.

Copyright © 1995 IOP Publishing Ltd.

9

Specialized Optical Spectroscopic Techniques for Astronomy

9.1 INTRODUCTION

Spectroscopy in the laboratory and in the observatory can sometimes be very closely related to each other, using the same instruments and techniques. More normally, however, spectroscopes designed for astronomy need to be optimized for that purpose (chapter 10), and different or additional techniques need to be used. We have already seen one major requirement, that of light gathering power or throughput. In terms of its utilization of the available light, the astronomical spectroscope must be as efficient as possible, because the sources are often very faint. Even those sources for which this would not seem to be a problem, such as the Sun, have been thoroughly studied in the past at high light levels, and current work uses very high spectral and/or angular resolutions so that light efficiency is again a fundamental requirement. This chapter looks at the other specialized requirements for spectroscopy as used in astronomy.

9.2 DETECTORS

Generally the same detectors can be used for astronomical spectroscopy as those used for direct astronomical imaging. The photographic emulsion is still to be found, especially in older instruments and for objective prism spectroscopy. However it is rapidly being replaced over the visible, near infrared and near ultraviolet by CCD detectors. These have a much higher quantum efficiency, even at the shorter wavelengths where fluorescent coatings have to be used. The image is fed directly into a computer where it may be easily processed (see later discussions). CCDs are, however, still relatively small, and so inappropriate for physically large spectra. Image Photon Counting Systems (IPCS) are also to be found and have the advantage of a continuous read-out of the image during exposure (see below). At wavelengths longer than about 1 μm, various array detectors based upon cooled semiconductor bolometers, some with charge

Copyright © 1995 IOP Publishing Ltd.

coupling read-out, are now in widespread use and have similar advantages to CCDs. In the ultraviolet, observations at wavelengths less than about 350 nm require the instrument to be lifted above the Earth's atmosphere. The main ultraviolet spectra to date have been obtained by the International Ultraviolet Explorer (IUE) spacecraft and the Hubble space telescope. The former uses an SEC vidicon TV camera as the detector within an echelle spectroscope operating down to 100 nm. The latter is discussed further in chapter 10.

9.3 GUIDING

One obvious difference between a laboratory and an astronomical source is that the latter is changing its position with time. Now all telescopes, except the cheapest ones aimed at the amateur market, are driven to counteract the effects of the Earth's rotation. So the changing position in the sky of the object under observation would not seem to matter for any spectroscope attached to or fed by such a telescope. Unfortunately the drives on most telescopes, including some of the largest and most modern, are inadequate to keep the stellar image centred on the entrance slit of the spectroscope, which may have a projected angular width on the sky of only 0.1″ or less. Even a drive which perfectly counteracted the Earth's rotation would not compensate for the changing refraction within the Earth's atmosphere as the zenith distance of the object altered, and so would fail to follow the object correctly throughout a long exposure. A system for correcting the imperfections in the telescope drive is therefore normally essential. This is generally known as guiding the telescope.

Guiding the telescope is normally essential for any purpose, not just for spectroscopy. For direct imaging, it is even more important, because any drift of the image will show up as a distortion (trail) of the image. In spectroscopy, drift will lead to longer exposures because the light from the object is not entering the spectroscope at all times, but without otherwise normally causing a deterioration of the spectral image. Guiding of a telescope is usually accomplished by viewing the object through a second telescope attached to the main one and adjusting the telescope's drive to keep the object centred on cross wires. This may be done visually, or there are numerous automatic systems fulfilling the same purpose. For very faint objects, off-set guiding may be necessary, when a brighter nearby object is used for guiding, and the guiding telescope off-set from the main telescope so that the object of interest remains centred in the main telescope. Alternative approaches include sampling a small portion of the field of view of the main telescope and guiding on an object in that region, or using a dichroic mirror to enable guiding to take place at one wavelength while the image is obtained at a different wavelength.

For some approaches to spectroscopy, such as multi-object slit spectroscopy, objective prism spectroscopy, or when the whole image of the object is accepted by the spectroscope, guiding has to be undertaken in the same manner as for

Copyright © 1995 IOP Publishing Ltd.

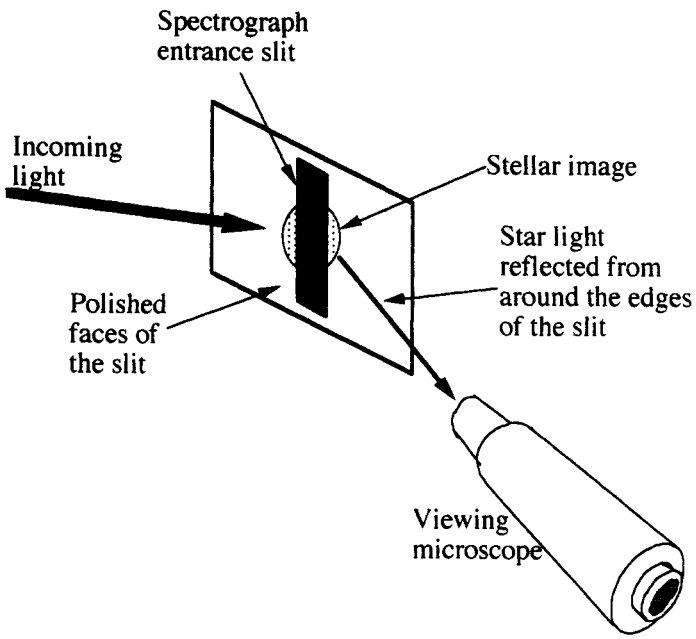

Figure 9.1 Use of a low power microscope to guide on the overspill for an image from the edges of the entrance slit to a spectroscope.

direct imaging. Much of the time, however, with single object slit spectroscopy, the image of the object, even if it is a star, will be larger than the entrance slit to the spectroscope. The overspill from the image may then be viewed via a low power microscope and used for guiding (figure 9.1).

9.4 WIDENING

The image of a star is not a point, nor even is it diffraction limited, because of spread caused by the Earth's atmosphere, arising from seeing, scintillation and twinkling. Nonetheless, a stellar image is normally only one or two seconds of arc across. If that image were to be guided to a fixed position on the slit, the resulting spectrum would be very narrow, and difficult to study. The spectrum is therefore normally deliberately widened to make it easier to analyse. This does, of course increase the required exposure in direct proportion to the increased width of the spectral image. Widening may be accomplished by setting the slit parallel to the diurnal motion of the object, and adjusting the telescope drive to be slightly too slow or too fast. The image will then drift slowly along the slit, and can periodically be reset to the other end of the slit using the drive's slow motion controls. More usually, however, the spectrum is widened using a separate device known as a rocker. This is a thick slab of glass with optically flat,

Copyright © 1995 IOP Publishing Ltd.

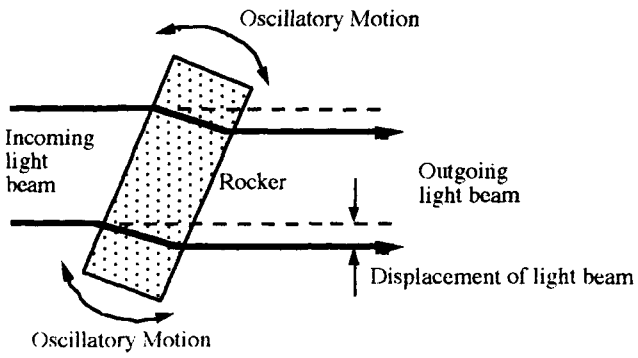

Figure 9.2 Use of a rocker to widen a spectrum.

parallel sides (figure 9.2). The slab is placed into the light beam and oscillated back and forth through a small angle. The varying displacement introduced into the beam by the slab then produces the required widened spectral image.

9.5 IMAGE DISSECTORS

The overspill from an image around the sides of the spectroscope entrance slit, though useful for guiding, represents a loss of light to the spectroscope and therefore results in an increased exposure. The light in the image can be more efficiently used through the use of an image dissector. This will also have the incidental benefit of widening the spectrum without requiring a separate mechanism. Image dissectors for modern spectrographs are made from a bundle of optical fibres. One end of the bundle is placed at the focus of the telescope and is matched in shape and size to the stellar image. The other end of the bundle feeds the spectroscope and is shaped to match its entrance slit (figure 9.3). Older devices used stacks of mirrors to the same end (figure 9.4).

9.6 DEKKERS

Most spectroscopes can change the available length of the slit by the use of an external stop. This will also usually have separate apertures to allow the comparison spectrum (see below) to be placed either side of the main spectrum without overlapping it. Such a stop is called a Dekker, and will usually have a range of lengths to allow for different objects and different observing conditions (figure 9.5).

Copyright © 1995 IOP Publishing Ltd.

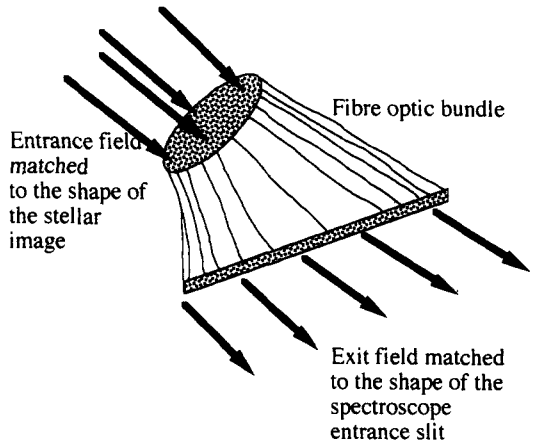

Figure 9.3 Image dissector formed from a fibre optic bundle.

9.7 LONG SLIT SPECTROSCOPY

The need for wideners and image dissectors is restricted to stellar spectroscopy. When an extended object is being observed, a much longer slit than normal can be used with advantage. The object needs to be guided to a fixed position on the slit to a high degree of accuracy. The final spectrum then gives, across its width, individual spectra for each portion of the object falling onto the slit (figure 8.20).

9.8 COMPARISON SPECTRA

9.8.1 Wavelength

From the earliest days of stellar spectroscopy it has been realized that a comparison spectrum would be needed to enable precise wavelength data to be obtained. Huggins made the first such attempt in 1868 when he compared the spectrum of Sirius with that of a hydrogen emission lamp. Few astronomical spectroscopes nowadays other than objective prisms and Fourier transform devices are without a means of producing such a spectrum. Indeed, one of the reasons for the use of a slit as the entrance aperture to a spectroscope is to facilitate the alignment of the comparison spectrum with the stellar (or other) spectrum.

The comparison spectrum is just an emission line spectrum, produced on or near the spectroscope from an emission lamp or an electric arc, the light from which is fed into the spectroscope. Usually the comparison spectrum is split and appears on either side of the spectrum of interest (figure 9.6). Measurement of the position of a spectral line (and hence its wavelength) on a photographic

Copyright © 1995 IOP Publishing Ltd.

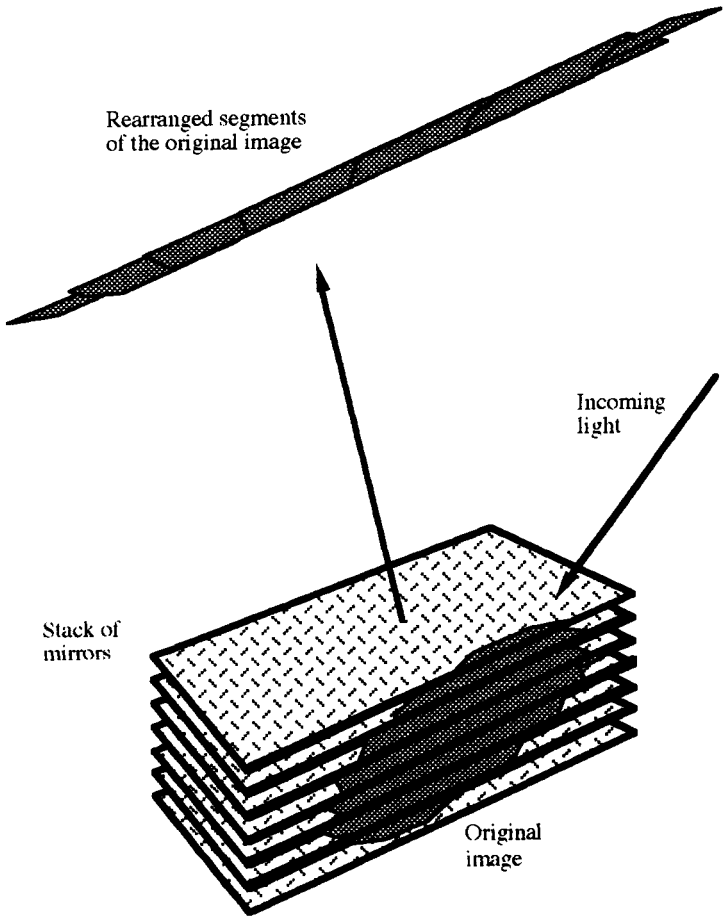

Figure 9.4 Image dissector formed from a stack of mirrors (Bowen image dissector).

spectrum is made by comparison with the emission lines' positions using a high precision travelling microscope (figure 9.7). For a CCD or equivalent image, a similar process takes place upon the electronic image stored within a computer using appropriate data reduction software. Since the wavelengths of the emission lines are known, the observed wavelengths of lines in the spectrum of interest can be found by interpolation. A good deal of care needs to be taken if accurate wavelengths are to be obtained. For example with a photographic spectrum, the measurements need to be repeated several times and the spectrum rotated through 180° between each pass to reduce setting errors. During a single such measurement, the screw driving the travelling stage should rotate in only one sense, in order to eliminate errors due to backlash. With both photographic and electronic images, care has to be taken when lines are asymmetrical, the dispersion is nonlinear, and so on. With such care, however,

Copyright © 1995 IOP Publishing Ltd.

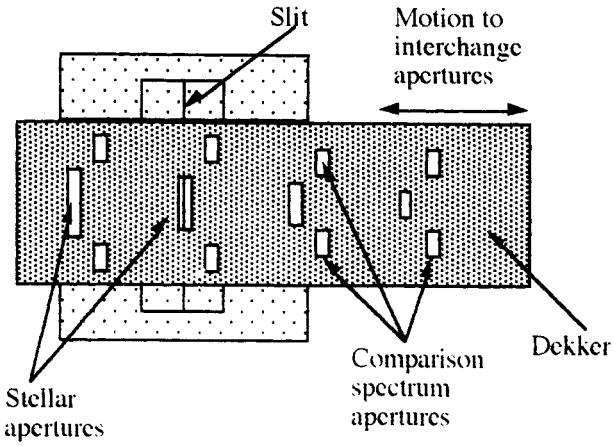

Figure 9.5 The use of a Dekker to limit the length of the entrance aperture of a spectroscope.

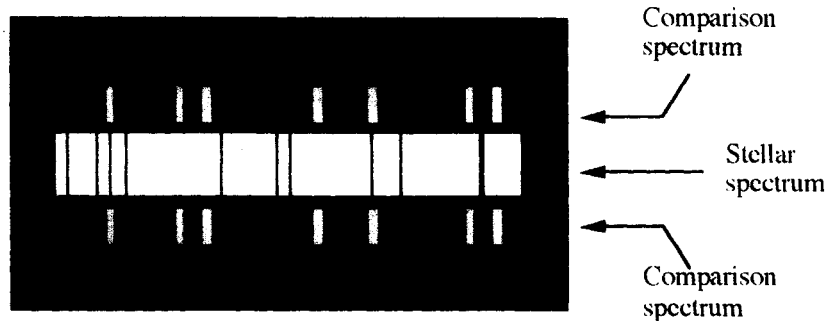

Figure 9.6 A stellar spectrum with a comparison spectrum.

observed wavelengths in the visible region may be determined typically to about one part in $30\,000/n$ (where n is the dispersion of the spectrum in nm mm^{-1}). This corresponds to being able to determine radial velocities through the Doppler shift to an accuracy of about $\pm 10n$ km s^{-1}.

9.8.2 Photometric

For detectors such as the CCD, which have a linear response, there is no problem in converting the detector output into relative spectral intensity. For nonlinear detectors, however, such as the photographic emulsion, a calibration curve is required to convert the signal into intensity. This curve is most usually obtained from a photometric comparison spectrum. A calibration curve, usually called the characteristic curve in the case of a photographic emulsion, may be obtained

Copyright © 1995 IOP Publishing Ltd.

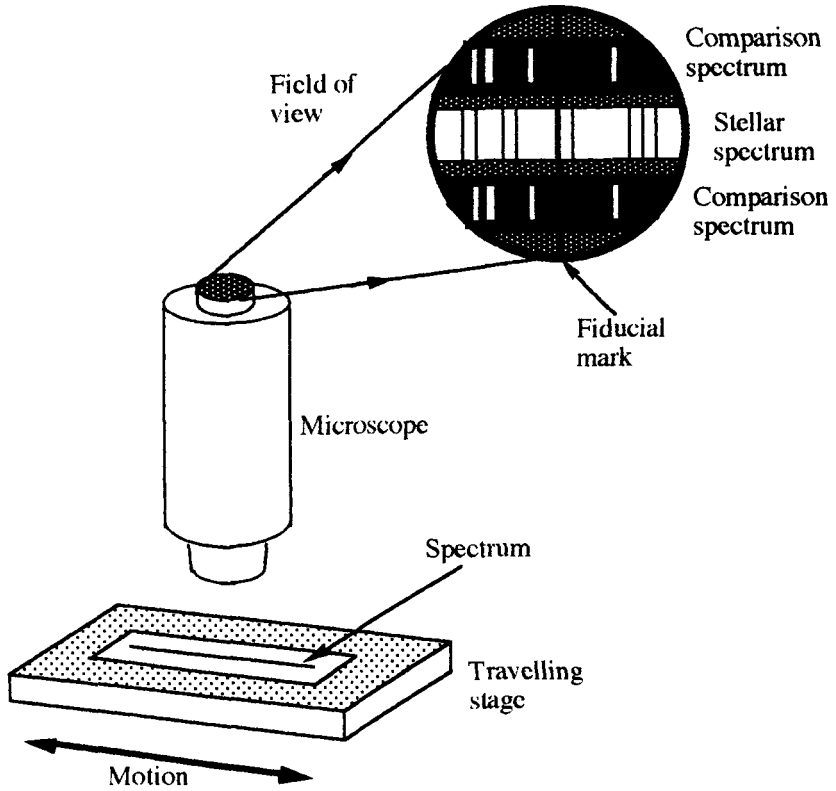

Figure 9.7 Use of the comparison spectrum to obtain the wavelengths of lines in the spectrum of interest.

in integrated light, but since the emulsion response may vary with wavelength, a calibration spectrum is more usual. This is a low dispersion spectrum with different sections across its width of different, and known, relative intensities (figure 9.8). From this a characteristic curve (figure 9.9) for the emulsion can be obtained, if necessary at a variety of wavelengths. Since the response of a photographic emulsion is highly variable with a number of factors, great care needs to be taken over the photometric calibration if accurate work is to be attempted. For example, the emulsion should be from the same batch as that used to obtain the spectrum, the exposure for the comparison should be of a similar length to that used for the main spectrum, the comparison and main spectra should be processed together etc. With such care errors in photographic spectrophotometry can be reduced to about 1%.

Although many of the difficulties of photography are avoided by the use of CCD and other array detectors with a linear response, there are still problems

Copyright © 1995 IOP Publishing Ltd.

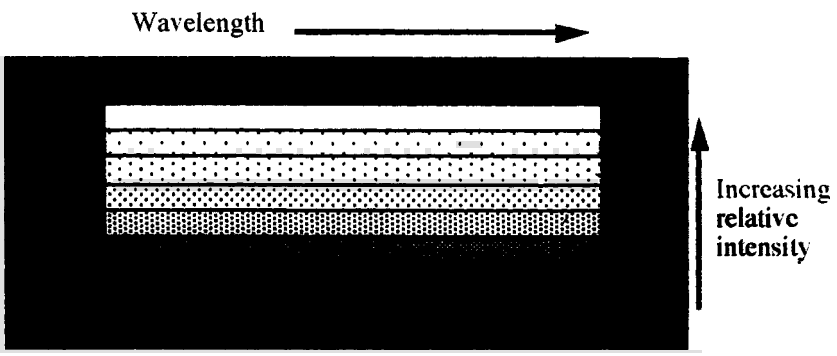

Figure 9.8 A photometric comparison spectrum—often emission lines may be added to facilitate identification of wavelengths along the spectra.

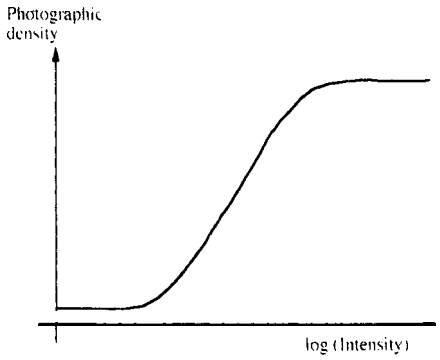

Figure 9.9 Schematic characteristic curve for a photographic emulsion.

to consider. The response of an individual pixel in an array may be linear, but the sensitivities of different pixels can vary. Furthermore there will be a dark signal (equivalent to the background fog on a photographic emulsion) and this may also vary from pixel to pixel. These effects have to be removed by taking exposures with similar lengths to that of the main exposure, firstly while observing a uniformly illuminated surface (flat fielding) to find the varying responses of the pixels, and secondly without any light incident onto the detector (dark frame subtraction) to reduce the effect of the dark signal. Since charge transfer in a CCD is not 100% efficient, a very bright part of the image will give spuriously high readings in adjacent pixels during the read-out process. This problem is of more importance in direct images where one or two stars may be very much brighter than the rest, but it is not negligible in spectrophotometry, especially when there are strong emission lines in the spectrum. There may also be 'bad" pixels, which have a poor response, very high dark signal or poor charge transfer, whose effects cannot be completely corrected by the flat fielding or dark frame subtraction. Then there are the effects of cosmic rays upon any

Copyright © 1995 IOP Publishing Ltd.

of the three exposures. A cosmic ray passing through the detector during an exposure will ionize a large number of the atoms, and the resulting electrons will be gathered up in the same way as those produced by the photons contributing to the image. One or two pixels around the point of entry of the cosmic ray will thus have an anomalously high signal. Such pixels can usually be recognized by careful inspection of the image, and as a partial remedy, the spurious signal replaced by the average of the eight surrounding pixels. Unfortunately, at the time of writing, correction of cosmic ray spikes is a highly tedious process which usually still has to be done by hand.

9.9 FLEXURE

The components of an astronomical spectroscope must be kept in their correct relative positions if the optical performance of the instrument is not to be degraded. With the exception of instruments placed at the Coudé focus of a telescope (or with an alt-az type mounting, at one of the the Nasmyth foci), the orientation of the spectroscope will vary as the telescope is pointed to different parts of the sky. The instrument must thus be designed so that the flexure arising from the changing gravitational loadings on it does not alter the positions and orientations of its components beyond the point at which the performance of the instrument would start to deteriorate. Early spectroscopes followed closely the design of the basic spectroscope (figure 8.14), and were attached in place of the eyepiece. They were therefore poorly supported and suffered badly from flexure. To reduce the problem the practice of reflecting the light beam from the telescope through 90° before incidence onto the spectroscope was soon adopted. The plane of the spectroscope is then the same as that of the primary mirror. A substantial mount for the spectroscope can thus be attached at several points to the equally substantial mounting for the primary mirror, reducing flexure to manageable proportions. Most modern spectroscope designs follow this practice, but care is still needed in their design to minimize the remaining flexure. The use of a folded design for the spectroscope, such as the Wadsworth (figure 8.18), can often be beneficial.

9.10 TEMPERATURE

Except for spectroscopes at the Coudé focus of a telescope, which may be kept in a temperature controlled room, most instruments are attached to the telescope, and so must be at or close to the ambient temperature. This is usually low and will vary throughout the night. The relative positions of the spectroscope's components can therefore alter through the thermal expansion or contraction of their supports. The use of suitable materials in the construction of the instrument, such as Invar can usually eliminate this difficulty. If any problems remain, then

Copyright © 1995 IOP Publishing Ltd.

the spectroscope itself can be heated to keep it at a constant temperature a few degrees above the ambient temperature. However, this practice can reduce the quality of the images through the production of convection currents near the telescope.

More seriously, the dispersing element may be affected by temperature changes. If it is a diffraction grating, then the use of zerodur, ULE or other low expansion material for its construction will reduce the problem almost to zero. The dispersion of a prism, however, is quite temperature dependent. If the spectral resolution is not to be degraded, then the temperature of the prism must be kept constant to about 0.1 K throughout the exposure. This can only be accomplished by the use of a precision thermostat and heating jacket.

9.11 EXPOSURES

The exposure required for a spectral image clearly depends upon the throughput, the widening and the detector sensitivity. The effects of these factors can be determined, at least approximately, in advance. However, spectroscopy, unlike much work in astronomy, can continue when the sky transparency is poor, or even through thin cloud. Furthermore, an exposure interrupted by cloud can be continued once the clouds have gone. It is even possible, given a stable enough spectroscope, to continue an exposure the following night; Edwin Hubble in his pioneering work on the red shift of the galaxies took exposures which extended over several complete nights. In such circumstances, it is clearly difficult to judge the correct exposure needed, and this is complicated further by the required exposure varying with the purpose for which the spectrum is needed. If strong emission lines are to be studied, then a much shorter exposure can be given than if faint absorption lines are of interest. Spectra from which radial velocities are to be measured can be of a lower signal to noise ratio than those from which line profiles are desired, and so on. For some purposes, Bowen's formula may be of use as a guide to the limiting magnitude for a spectroscope in which the star's image is larger than the slit width:

$$m = 12 + 2.5 \log_{10} \left(\frac{s D_1 T_D g q t (\mathrm{d}\lambda/\mathrm{d}\theta)}{f_1 f_2 H \alpha} \right) \tag{9.1}$$

where m is the faintest B magnitude which will give a usable spectrum in an exposure of t seconds, s is the projected slit width, D_1 is the diameter of the light beam passing through the dispersing element, T_D is the diameter of the telescope objective's, g is the optical efficiency of the system (i.e. the ratio of the usable light at the focus of the spectroscope to that incident upon the telescope)—typical values would be in the range 0.1 to 0.5, q is the quantum efficiency of the detector, $\mathrm{d}\lambda/\mathrm{d}\theta$ is the dispersion of the spectrum, f_1 and f_2 are the focal lengths of the collimator and imaging element respectively, H is the

Copyright © 1995 IOP Publishing Ltd.

height of the spectrum and α is the angular size of the stellar image at the slit. With some detectors, such as the IPCS, there can be a continuous display of the spectrum throughout the exposure, and it can simply be continued until the desired signal to noise ratio is achieved. With CCDs, it is easy to add separate exposures together, though this does increase the read-out noise. If an image is read out, and is found to be underexposed, then a second or third exposure can be added, again until the desired signal to noise ratio is achieved. For photographic spectroscopes it is less easy, though not impossible, to add images together.

Some instruments have an integrating exposure meter. This samples a small part of the spectrum and integrates the signal. The exposure can then be stopped when an adequate exposure has been given. Much of the time, however, no such devices are available, and the observer has to act as his or her own exposure meter, and estimate appropriate allowances for sky transparency, cloud etc.

Copyright © 1995 IOP Publishing Ltd.

10

Examples of Optical Spectroscopes

10.1 INTRODUCTION

In this chapter some examples of actual astronomical spectroscopes are examined. The danger of reviewing current instruments in a book, of course, is that they may be out-of-date by the time the book appears, or soon afterwards. Nonetheless it is still worth looking at current instruments for several reasons. Firstly, the development of spectroscopes has been occurring for well over a century, and current examples are therefore close to the limits of what can be achieved until some very radical new development occurs. Secondly, and partly because of the first reason, individual instruments tend to remain in use for long periods with only minor modifications. The author for example has used spectroscopes constructed in the early 1950s that are still to be found on some major telescopes. Finally, and again because of the first reason, even if new instruments do come into use, they are unlikely to be fundamentally different from existing designs.

No examples of prism-based slit spectroscopes have been included, because, whilst some such instruments may remain in use, it is now highly unlikely that any new spectroscope for a major telescope would be designed purely using a prism as the dispersing element. Designs utilizing objective prisms and grisms (combined gratings and prisms) are covered, however.

10.2 A SMALL BASIC SPECTROSCOPE

A simple grating-based slit spectroscope, designed for use on a small telescope and primarily intended for teaching purposes, is shown in figure 10.1. Its optical layout (figure 10.2) clearly shows all the elements of the basic spectroscope (figure 8.14), and its parameters are listed in table 10.1. Such an instrument on a moderate telescope is capable of providing reasonable spectra of the brighter stars, sufficient for spectral classification, monitoring spectrum variables, for element identification etc.

Copyright © 1995 IOP Publishing Ltd.

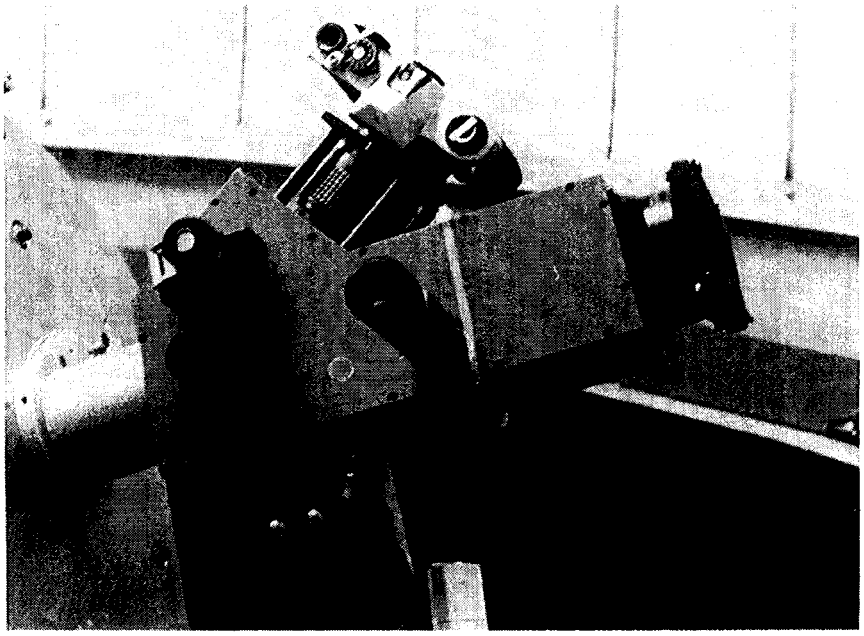

Figure 10.1 A simple basic spectroscope attached to a 0.4 m telescope (Optomechanics™ model 10C).

10.3 A CONVENTIONAL CASSEGRAIN SPECTROSCOPE

The workhorse of spectroscopy is a conventional grating-based slit spectroscope attached to the Cassegrain focus of a large telescope. Its size is limited by the space available at the back of the main mirror mounting of the telescope, and its weight is limited by the strength of the telescope's tube and mounting. There are numerous examples of this type of instrument, only one of which, ISIS, is discussed here.

ISIS (Intermediate dispersion Spectroscopic and Imaging System) is one of the most frequently used spectroscopes for the 4.2 m William Herschel telescope on La Palma. It is fixed at the Cassegrain focus, and is reasonably representative of this type of instrument except that it is a dual system. It uses a dichroic mirror to split the incoming light into a red beam and a blue beam, each of which is then separately directed into a spectroscope optimized for those wavelengths (figure 10.3). The slit is common to both beams and can be varied from 50 μm to 5 mm in width (0.2″ to 23″ on the sky). It is followed by a dichroic mirror which separates the incoming white light into the blue beam (reflected) and the red beam (transmitted). Several mirrors are available to give the separation point at various wavelengths in the range 450 to 550 nm. The parameters of the system are listed in table 10.2.

Copyright © 1995 IOP Publishing Ltd.

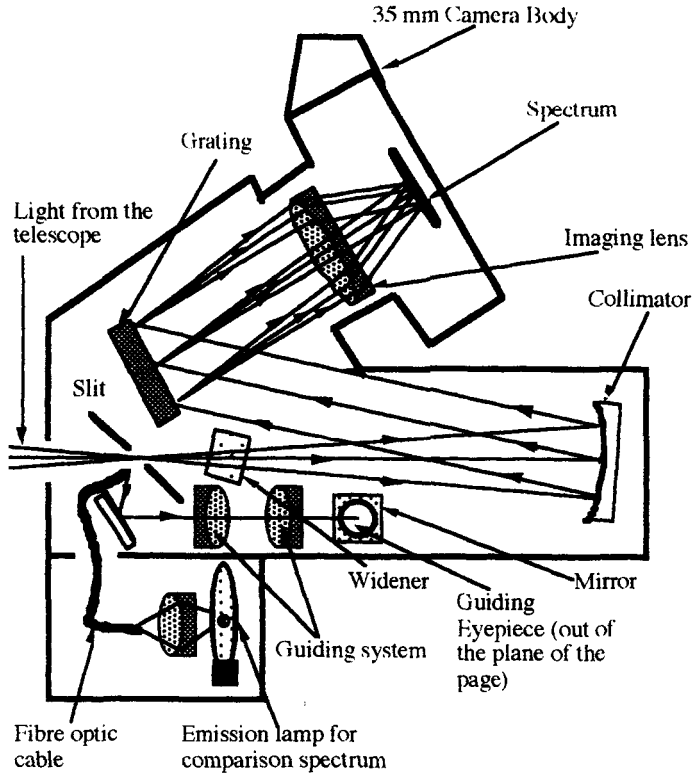

Figure 10.2 The optical layout of the Optomechanics 10C spectroscope.

Table 10.1 Parameters of the Optomechanics model 10C spectroscope.

Slit width	50 or 100 μm
Collimator	Focal length 225 mm, $f9$
Gratings	Reflection grating, 600 lines per mm, blazed for 500 nm in the first order, dispersion 12 nm mm^{-1}. Spectral resolution 500 or 1 000
	Reflection grating, 1200 lines per mm, blazed for 500 nm in the first order dispersion 6 nm mm^{-1}. Spectral resolution 1 000 or 2 000
Imaging element	Focal length 135 mm, $f2.8$
Comparison spectra	Fe/Ar or Hg/ Ne

10.4 TRANSMISSION GRATING SPECTROSCOPES

The use of a transmission grating is much less common than the use of a reflection grating in astronomical spectroscopes, but some examples do exist.

Copyright © 1995 IOP Publishing Ltd.

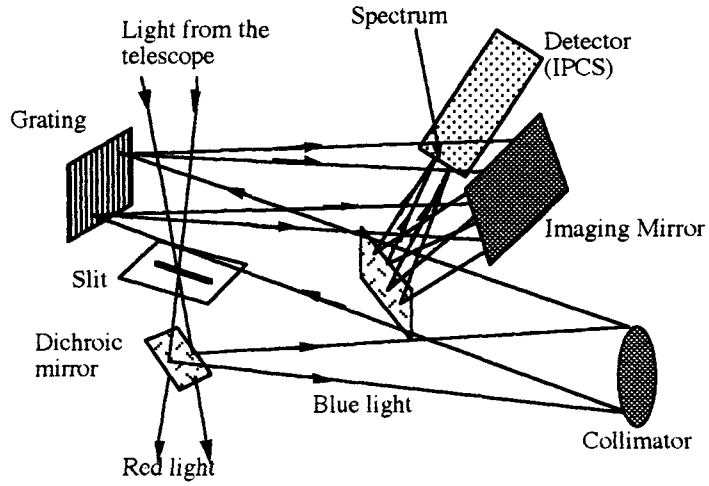

Figure 10.3 Schematic optical layout of the blue arm of ISIS; the red arm is similar and placed underneath the blue arm.

Table 10.2 Parameters of ISIS.

Slit width	50 μm to 5 mm
Collimators	Each of focal length 1.65 m, producing beams 150 mm in diameter
Gratings	Various reflection gratings, each 154× 206 mm in size, with from 150 to 2400 lines per mm, giving dispersions from 0.8 to 12 nm mm^{-1}. The blaze wavelengths range from 360 to 720 nm, and the spectral resolution from 800 to 7000
Imaging elements	Each of focal length 500 mm, f1.0
Detectors	For the blue beam, an IPCS with a CCD as the detector; for the red beam, a CCD
Comparison spectra	Cu/Ar, Cu/Ne, Th/Ar, or a continuum
Guiding	An auto-guider, a TV system for viewing the slit, or a blind offset from a brighter guide star
Optional elements	A multislit or fibre optic feed, a cross disperser, a polarizer, neutral density and colour filters (the latter to eliminate overlap between orders)

Thus, ISIS (above) has a faint object spectroscope (FOS) attached to it as a third arm, which relies on a transmission grating for its main dispersion. In the FOS, the grating is cemented to a direct vision prism which acts as a cross disperser (such a combination is known as a Grism, see below), in order to separate the first and second order spectra.

Here we look at another example of a transmission grating-based spectroscope, the Low Dispersion Survey Spectroscope (LDSS). This instrument

Copyright © 1995 IOP Publishing Ltd.

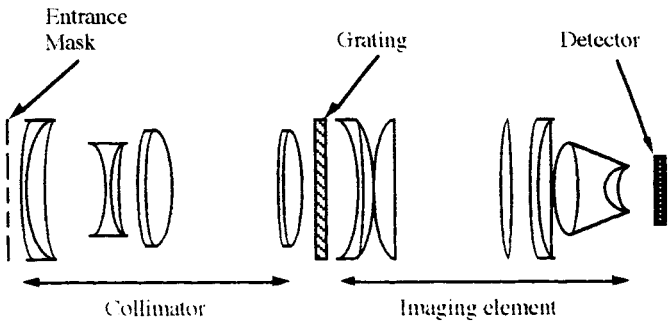

Figure 10.4 Optical layout of LDSS (schematic).

Table 10.3 Parameters of LDSS

Slits	Up to 100 slits 50 μm wide (plus long slit, no slit and direct options)
Gratings	Two, giving dispersions of 16.5 and 87 nm mm^{-1} with spectral resolutions of about 400 and 70, respectively
Detectors	IPCS or CCD
Spectral range	370 to 750 nm
Throughput	About 60%

is designed to obtain the spectra of up to 100 objects simultaneously at low dispersions, and down to about magnitude 23 when attached to the 3.9 m Anglo–Australian telescope (AAT). For greatest efficiency, the spectroscope uses lenses throughout in order to avoid the obstructions caused by the secondary mirrors in reflecting systems. All the lenses have multilayer coatings to reduce reflection losses. The entrance aperture can take several forms. In the principal mode of operation the entrance aperture is a mask with up to 100 precisely positioned slits to match the positions of objects in the field of view. Each mask is unique and has to be made specifically for each field of view observed with the instrument. Because of the width of the entrance field, the optics of this spectroscope need to have their aberrations corrected over a much wider field of view than those for normal spectroscopes, and this requirement is a second reason for the choice of transmission optics. The optical layout of the spectroscope is shown in figure 10.4, and its parameters listed in table 10.3.

10.5 COUDÉ SPECTROSCOPES

The weight and size constraints imposed on a Cassegrain spectroscope are considerably relaxed for a spectroscope placed at the Coudé or Nasmyth focus of a telescope. The problems of flexure are also eliminated or much reduced, because the spectroscope does not change its attitude as the telescope moves.

Copyright © 1995 IOP Publishing Ltd.

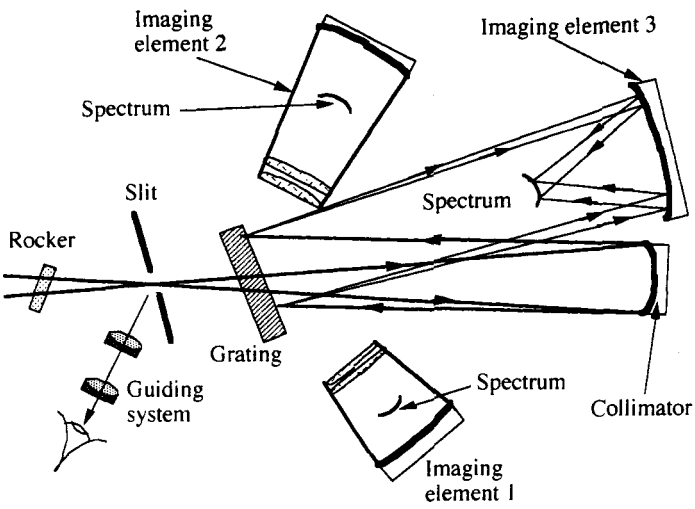

Figure 10.5 The Coudé spectroscope on the 1.52 m telescope of the European Southern Observatory (ESO) at La Silla, showing the light paths to the third imaging element.

The image from the telescope can often be brought into an area separate from the main observatory through an optical window. This area can then be temperature controlled to suit either the operators or the spectroscope (or both), reducing the problems due to temperature variations and improving observing conditions. Coudé and Nasmyth spectroscopes can thus be much larger, have higher dispersion and/or resolution, and be less robustly constructed than Cassegrain spectroscopes. The disadvantages of a Coudé or Nasmyth focus are that the image rotates as the telescope tracks the object across the sky, and that the focal ratio is usually very long.

As an example of this type of instrument, we look at the Coudé spectroscope on the 1.52 m telescope of the European Southern Observatory (ESO) at La Silla. This instrument uses reflection gratings in a conventional design. Three separate imaging elements are incorporated into the one mounting and can be selected by changing the angle of the grating. The optical layout of the spectroscope is shown in figure 10.5, and its parameters listed in table 10.4. Note the much better dispersion and spectral resolution possible with this instrument compared with ISIS.

10.6 A GRISM-BASED SPECTROSCOPE

The combination of a transmission grating and a prism, known through an unpleasant contraction as a Grism, can sometimes offer advantages over a pure grating-based system. This is particularly the case when a high efficiency instrument is required for the study of the spectra of very faint objects. The Faint

Copyright © 1995 IOP Publishing Ltd.

Table 10.4 Parameters of the Coudé spectroscope on the ESO 1.52 m telescope.

Slit width	50 to 300 μm (0.2″ to 1.3″ on the sky)
Collimators	Focal length 6 m, $f30$
Gratings	Various reflection gratings, each about 200 × 300 μm in size, with from 770 to 1500 lines per mm, giving dispersions from 0.26 to 3.1 nm mm^{-1}. The blaze wavelengths range from 410 to 750 nm, and the spectral resolution from about 10 000 to 100 000
Imaging elements	Three, of focal lengths 0.41, 0.67 and 0.25 m, and focal ratios 1.0, 1.6 and 2.2, respectively
Comparison spectra	Fe, Ne , Ar, or a continuum
Options	A field rotator to compensate for the rotation of the field of view due to the tracking motion of the telescope, and an exposure meter which uses the light blocked by the plate holders

Object Spectroscope (FOS) of the William Herschel telescope is an example of such a spectroscope. It is designed for low dispersion spectroscopy over a wide spectral range, and achieves this by using both the first and second order spectra produced by the grating and separating them by a direct vision prism acting as a cross disperser (figure 10.6). The two spectra overlap to give continuous coverage from about 350 nm to 970 nm. The optical layout of the instrument is shown in figure 10.7 and its parameters in table 10.5. In order to improve efficiency, the number of surfaces is kept to a minimum, so no collimator is used, and the imaging element is a Schmidt camera. The instrument is attached to the ISIS spectroscope (see above) at the Cassegrain focus of the WHT, and shares some of its components. It can be used simultaneously with the blue arm of ISIS.

10.7 MULTI-OBJECT SPECTROSCOPES

The LDSS (see above) can obtain the spectra of up to 100 objects simultaneously and is one example of this class of instrument. It requires, however, a mask containing slits positioned to match the objects to be studied and this has to be made up for every observation. A more flexible system, which also does not require the large field required for the optics of LDSS, is to use fibre optic cables to bring the light from the stars in the field of view to the slit of the spectroscope. The fibre optic cables can be moved around the image plane to intercept the light from the required objects easily compared with the labour required for machining a precision mask. A number of such instruments have been brought into use in the last decade, of which the ESO's MEFOS (Meudon–ESO Fibre Optics Spectroscope) is taken as an example.

MEFOS is positioned at the prime focus of the ESO's 3.6 m telescope, and feeds a spectroscope at the Cassegrain focus. The fibre optic cables are 21 m

Copyright © 1995 IOP Publishing Ltd.

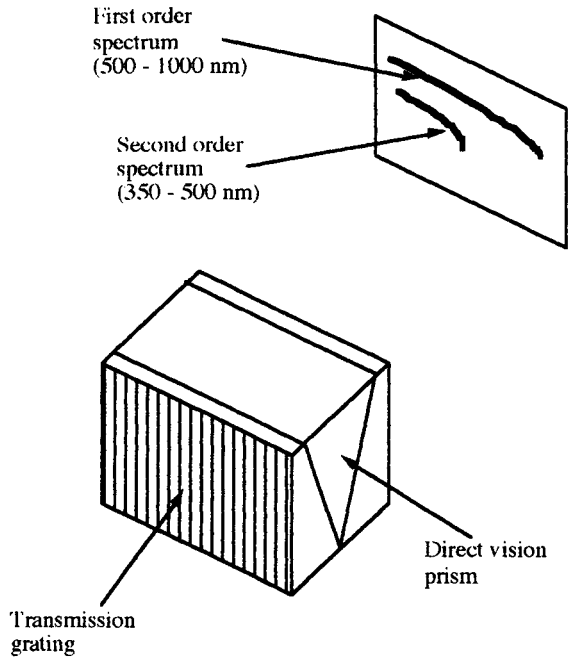

Figure 10.6 The Grism of the William Herschel Faint Object Spectroscope.

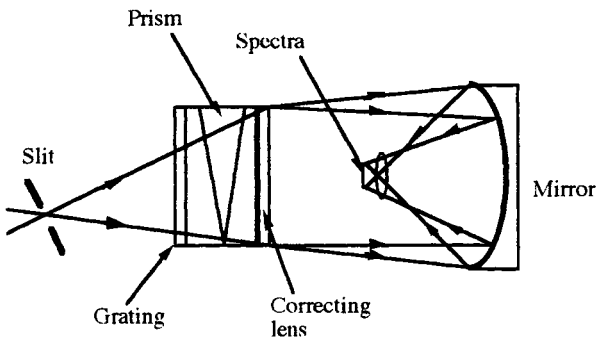

Figure 10.7 The optical layout of the William Herschel Faint Object Spectroscope.

long and the total light losses introduced by MEFOS are estimated at about 20%. It can observe up to 30 objects simultaneously over a 1° field, and the fibre optics are moved to the calculated positions of those objects by computer-controlled positioning arms (figure 10.8). Three fibres are carried by each arm. One intercepts the light from the object to be observed, the second the light from the nearby sky to provide background subtraction; these two fibres each cover

Copyright © 1995 IOP Publishing Ltd.

Table 10.5 Parameters of FOS.

Slit	50 μm to 5 mm or 0.2″ to 23″ on the sky (shared with ISIS to which the FOS is attached, see above). Two slits are used in order to obtain a sky background simultaneously with the stellar spectrum
Collimator	None. The system accepts the f11 beam from the focus of the WHT directly
Grating	A plane transmission grating, blazed at 730 nm, with 150 lines per mm cemented to a direct vision prism as a cross disperser. The first order spectrum covers from 460 to 970 nm at a dispersion of 40 nm mm^{-1}. The second order spectrum covers from 350 to 490 nm at a dispersion of 20 nm mm^{-1}. The spectral resolution is about 500 in the first order and 1000 in the second order
Imaging element	An f1.4 Schmidt camera accepting an f11 incoming beam
Detectors	A CCD detector
Throughput	17% including telescopic and atmospheric losses

2.6″ on the sky. The third (actually a fibre optic bundle covering 36″ × 36″ on the sky) provides a separate direct image for positioning purposes. The f3 beam from the telescope feeds with high efficiency directly into the fibres. No front lenses or prisms are required as can be the case with some other comparable systems. To start an observation, the positioning arms first move the imaging bundles to the calculated positions of the desired objects (figure 10.9). The actual alignments of the imaging bundles with respect to their targets are then found by taking a short direct exposure. The positions of the arms are corrected to centre the targets and are then offset to align the images precisely onto the spectroscopic fibres. The light from the objects is then fed into a spectroscope for the actual exposure. Guiding is accomplished by setting one of the imaging fibres onto a convenient bright star in the field. The system can be used with most spectroscope designs provided the collimator is matched to the f3 beam emerging from the fibre optics.

10.8 AN ECHELLE GRATING SPECTROSCOPE

An echelle grating can give high dispersions and spectral resolutions, but in order to achieve these it usually has to be used at high spectral orders and so must be allied to a cross disperser. The resulting spectrum is then a two-dimensional array of short segments of spectra. A major consideration in the design of an echelle spectroscope is therefore to ensure overlap at the ends of successive segments so that the whole spectrum may be reconstituted. As an example of an echelle grating based spectroscope, we look at those on board the International Ultraviolet Explorer (IUE) spacecraft.

The IUE spacecraft is designed specifically to obtain spectra in the ultraviolet.

Copyright © 1995 IOP Publishing Ltd.

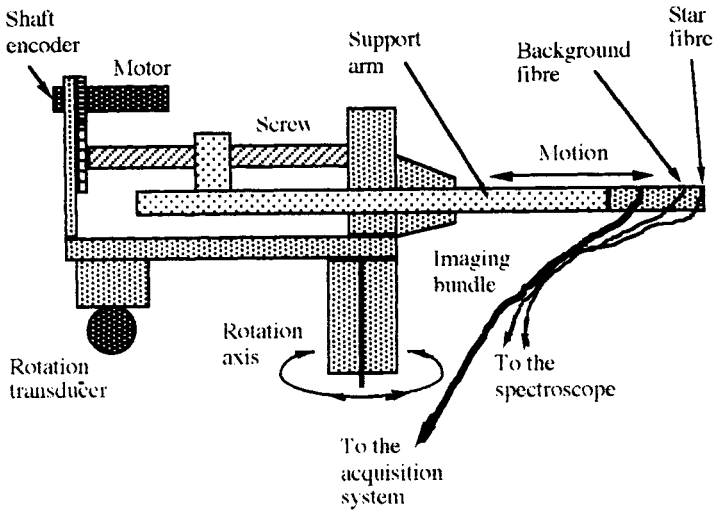

Figure 10.8 Schematic side view of a position arm from MEFOS.

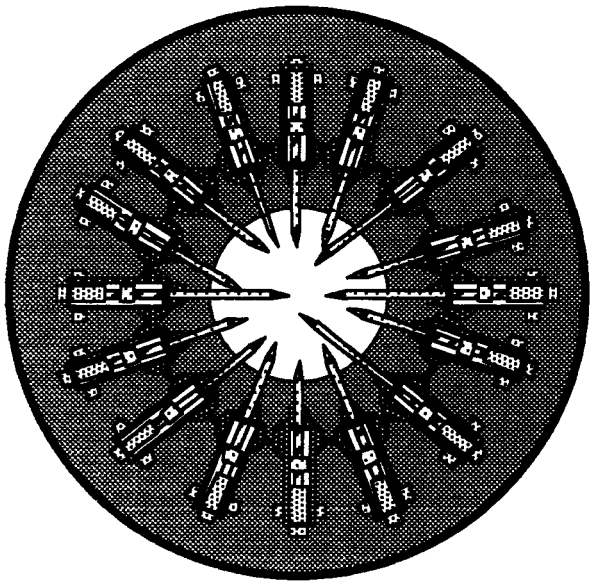

Figure 10.9 Schematic view of MEFOS (only 16 of the 30 positioning arms actually shown).

It carries a 0.45 m, f15 Ritchey–Chretien telescope which has two spectroscopes at its 'Cassegrain' focus. These spectroscopes cover the wavelength ranges

Copyright © 1995 IOP Publishing Ltd.

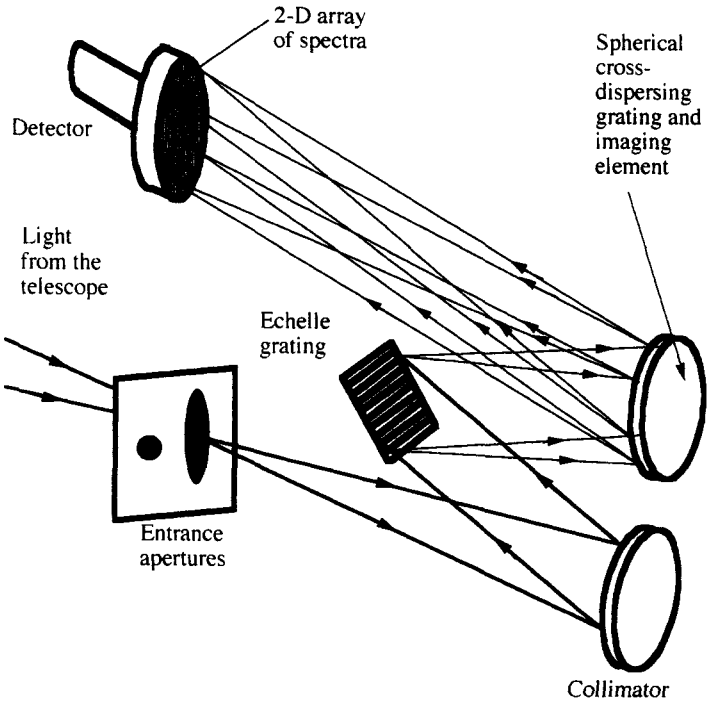

Figure 10.10 The optical layout of one of the spectroscopes on board the IUE spacecraft.

115–195 nm and 190–320 nm. Both spectroscopes use an echelle grating as the primary disperser, and then a low dispersion spherical grating which acts both as the cross disperser and as the imaging element (figure 10.10). The parameters of the spectroscopes are listed in table 10.6. An example of a high dispersion spectrum from IUE is shown in figure 8.10.

10.9 INFRARED SPECTROSCOPES

Full coverage of the infrared region requires that the spectroscope be lifted above most or all of the Earth's atmosphere, and balloon, rocket and spacecraft-borne spectroscopes have been so used. The atmosphere does, however, have some gaps in its near-infrared absorption spectrum, allowing radiation to reach the surface. Thus ground-based instruments, especially those working from dry sites at high altitude, can still be of use. The main difference between such an infrared spectroscope and those operating in the visible, apart from the obvious requirement for the optical components to transmit or reflect at the desired wavelengths, is the need to reduce background noise. Since the components of the telescope and spectroscope themselves can be emitting radiation in the

Copyright © 1995 IOP Publishing Ltd.

Table 10.6 Parameters of the IUE spectroscopes.

Entrance aperture	100 μm circular or 330 × 660 μm elliptical (3″ or 10″ × 20″ on the sky)
Collimators	Each off-axis paraboloids of focal length 1.89 m
Gratings	Short wavelength echelle grating; 102 lines per mm covering the range 115–195 nm over spectral orders 66 to 125 at a dispersion ranging from 0.086 to 0.137 nm mm^{-1}, and a spectral resolution of 12 000. The spherical grating used as a cross disperser has 313 lines per mm, and a focal length of 0.69 m.
	Long wavelength echelle grating; 63 lines per mm covering the range 190–320 nm over spectral orders 72 to 125 at a dispersion ranging from 0.125 to 0.2 nm mm^{-1}, and a spectral resolution of 13 000. The spherical grating used as a cross disperser has 200 lines per mm, and a focal length of 0.69 m. In both spectroscopes a plane mirror can be inserted in front of the echelle grating, leaving the cross disperser as the only dispersing element and giving a low dispersion mode to the system. The dispersions and spectral resolutions are then respectively 6 nm mm^{-1} and 250 for the short wavelength spectroscope, and 6 nm mm^{-1} and 300 for the long wavelength spectroscope
Detectors	An UV–optical converter followed by an SEC–vidicon television camera
Throughput	About 2% for the short wavelength spectroscope and from 0.45% to 4% for the long wavelength spectroscope (including the losses in the telescope)

infrared, this requirement usually translates into cooling some or all of the instrument. The ESO's infrared spectrometer, IRSPEC, is chosen as an example of this class of instrument.

This instrument was designed for use initially at the Cassegrain focus of the ESO's 3.6 m telescope, and later at the Nasmyth focus of the 3.5 m New Technology Telescope (NTT). The optical layout of the spectroscope is of relatively conventional design (figure 10.11). It is, however, encased in a vacuum chamber and cooled with continuously circulating liquid nitrogen to about 80 K, with the detector further cooled to 50 K by a thermal finger from solid nitrogen in a cryostat inside the main vacuum chamber. The optical components are mounted on an uncooled support from which they are thermally isolated, so that they may be aligned at room temperature and then cooled without going out of alignment. The parameters of the spectroscope are listed in table 10.7.

10.10 SPACECRAFT-BORNE SPECTROSCOPES

Any spectroscope operating at wavelengths shorter than about 350 nm, in the longer infrared or microwave regions, has to be lifted above the Earth's

Copyright © 1995 IOP Publishing Ltd.

Table 10.7 Parameters of IRSPEC

Aperture	800 μm (6″ on the sky)
Collimator	A 100 mm, f7.4 off-axis paraboloid
Gratings	Various reflection gratings, each 120 × 150 mm in size, covering the wavelength range 1–5 μm at spectral resolutions from 1 000 to 2 500
Imaging element	An off-axis, f2 Pfund-type camera
Detectors	A cooled infrared array
Comparison spectra	Ne or Kr
Guiding	Via a TV system for viewing the slit

atmosphere to make its observations. In other parts of the spectrum, being above the atmosphere can be a considerable benefit through the absence of absorption bands. At any wavelength, the observations will gain from the elimination of atmospheric image degradation (seeing), the reduction in background light, the absence of clouds and the reduction in gravitational loading. Thus many spectroscopes have been mounted onto spacecraft, rockets, balloons or very high flying aircraft to gain some or all of these advantages. One such spectroscope, that on board the IUE spacecraft, has been discussed in detail already. Here we look at another example, the Goddard High Resolution Spectroscope (GHRS) on the Hubble Space Telescope (HST).

The GHRS is designed to produce high quality spectra over the wavelength range 110–320 nm. The limits are due to the magnesium fluoride coatings on the telescope's optics and to the detector limits, not to the design of the spectroscope. Its optical layout (figure 10.12) is conventional but with a large number of options. Thus it may be used with conventional reflection gratings and a separate imaging mirror, or with an echelle grating and a concave cross-dispersing grating which also acts as the imaging element. Three of the conventional gratings are holographic and one ruled. The instrument also has two channels covering the ranges 110–180 nm and 115–320 nm. Exposures can be as short as 50 ms, and repeated rapidly for observation of changing sources. The longest single exposure is limited to about five minutes by motions within the instrument arising from the Earth's magnetic field. Longer exposures must be made by adding together several shorter exposures. Most objects are visible for about 45 minutes between 'rising' and 'setting' due to the spacecraft's orbital motion. Thus the longest exposures must extend over several orbits. The parameters of the instrument are listed in table 10.8.

10.11 FABRY–PÉROT SPECTROSCOPES

Fabry–Pérot etalons are most likely to be found in an observatory in use as narrow band filters; their use in spectroscopes is more usual in laboratory-

Copyright © 1995 IOP Publishing Ltd.

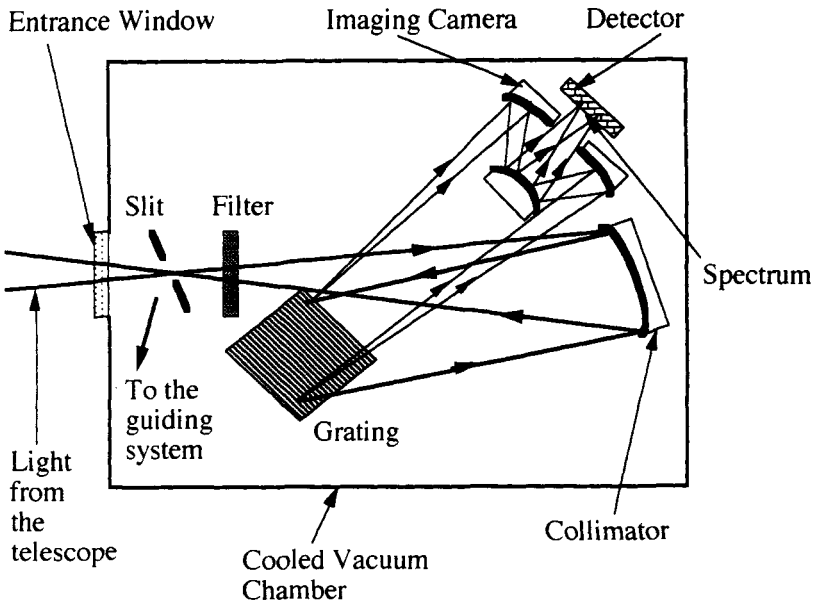

Figure 10.11 Schematic optical layout of IRSPEC.

Table 10.8 Parameters of GHRS.

Apertures	72 and 560 μm (0.22″ and 1.74″ on the sky)
Collimator	A 80 mm f23 off-axis paraboloid
Gratings	Four conventional reflection gratings, used in the first order, with from 600 to 6 000 lines per mm, covering the wavelengths from 110 to 320 nm at spectral resolutions from 2000 to 25 000. An echelle grating with 316 lines per mm, used at orders from 17 to 53, over the same wavelength range, at a spectral resolution of 80 000. Two cross-dispersing concave gratings with 195 and 86 lines per mm, and focal lengths of 1.46 and 1.34 m
Imaging elements	Two alternative concave mirrors, 84 mm, f17 and 86 mm, f15.7, respectively
Detectors	Two Digicon detectors, one with a caesium iodide photocathode and lithium fluoride window, detecting only wavelengths shorter than 180 nm, and the other with a caesium telluride photocathode and magnesium fluoride window for longer wave detection

based instruments. Nonetheless, there are a few astronomical examples, and we consider here TAURUS, which additionally serves as an illustration of the desperation with which acronyms may be sought since its name is derived from the 'Taylor–Atherton Uariable-resolution Radial Uelocity System'!

Copyright © 1995 IOP Publishing Ltd.

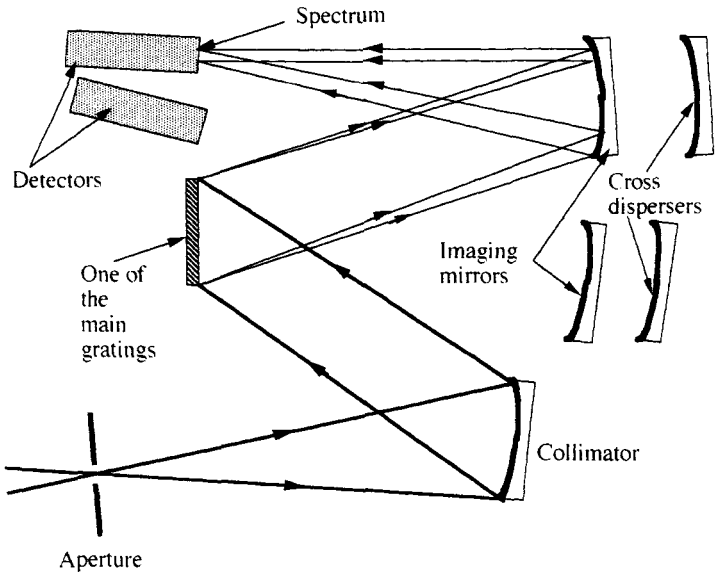

Figure 10.12 Schematic optical layout of the GHRS, showing just one of the many light paths that it is possible to select.

The second generation TAURUS-2 instrument was designed for the 3.9 m Anglo–Australian Telescope (AAT). It is intended primarily for mapping the velocities of emission-line objects. The instrument has the etalon placed in a collimated beam from the telescope. A filter isolates one of the transmitted orders, and the beam is re-imaged. The final image, detected by an IPCS or CCD detector, is thus pseudo-monochromatic. That is, at any point in the image the radiation covers only a very narrow waveband, but because of the changing angle of incidence onto the etalon, the centre of that waveband changes over the image. In operation, the spacing of the etalon is varied by piezoelectric spacers to scan its transmitted wavelength over the free spectral range of the order being used. Up to a hundred images may be observed. An image-cube is then reconstructed from the data with the image in two of the dimensions and the spectral dispersion orthogonal to this. The wavelengths across a single image have to be appropriately binned to allow for their variation over the field of view. The order to be observed is selected to be centred on one of the emission lines in the object. The velocities within the object observed can then be obtained from the change in the wavelength of the chosen emission line. The optical layout is shown in figure 10.13 and its parameters in table 10.9.

10.12 FOURIER TRANSFORM SPECTROSCOPES

These are even more rarely found in astronomy than Fabry–Pérot instruments, but they do occur in a few very specialist applications. Two of these few

Copyright © 1995 IOP Publishing Ltd.

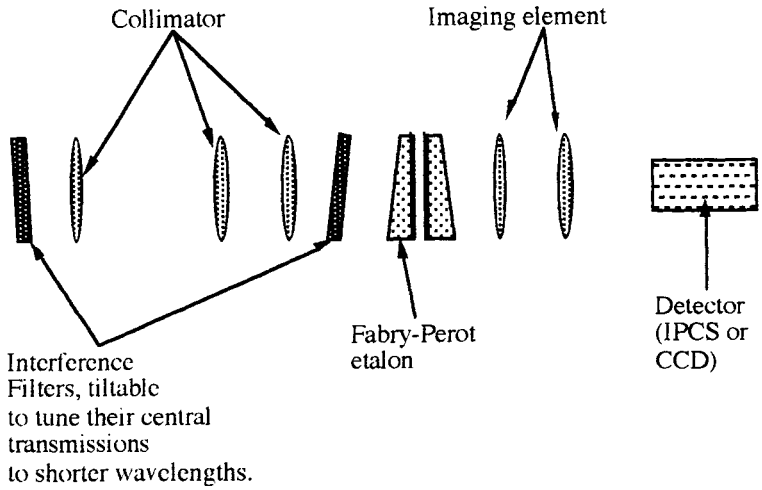

Figure 10.13 Schematic optical layout of TAURUS.

Table 10.9 Parameters of TAURUS.

Field of view	9 minutes of arc
Collimator	60 mm $f11$ lens system
Etalons	Various, with gaps ranging from 15 to 500 μm, velocity resolutions ranging from 7 to 150 km s^{-1}, spectral resolutions ranging from 500 to 60 000, spectral orders typically a few hundred, and covering the range 360 to 945 nm
Imaging elements	Two, with 61 mm, $f2.1$ and 61 mm, $f4$ lenses
Throughput	Up to 30%

examples are the Fourier Transform Infrared Spectroscopes (FTIS) on board the two Voyager spacecraft. The instruments were used, amongst other purposes, to identify compounds in Jupiter's, Saturn's and Titan's atmospheres, and to determine the composition of Martian clouds. The instruments had to be kept at 200 ± 0.5 K, and were therefore provided with thick thermal shielding and three sets of heaters, as well as radiators for cooling. They also had to be radiation-hardened in order to survive passage through Jupiter's radiation belts. The optical layout is shown in figure 10.14. In addition to the main Fourier transform spectrometer (FTS) shown in figure 10.14, the instrument had a separate reference FTS observing the 585 nm line from an on-board neon source, and the spherical dichroic mirror allowed visible light through to a separate radiometer.

Copyright © 1995 IOP Publishing Ltd.

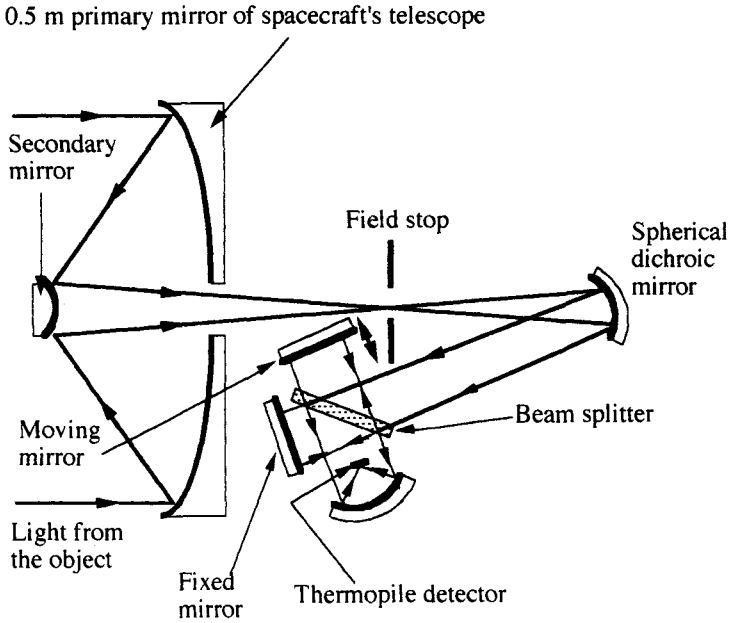

Figure 10.14 Optical layout of the FTIS on board the Voyager spacecraft.

10.13 OBJECTIVE PRISM SPECTROSCOPES

The simplest spectroscope of all is the objective prism. This (chapter 8) is just a large, low angle prism covering the whole aperture of the telescope, so that every object on the image is represented by its spectrum. Objective gratings can also be found, but these are for assisting photometric and astrometric measurements, not for spectroscopic purposes. Some variants of objective prisms replace the single large prism with a matrix of co-aligned smaller prisms in order to reduce costs, but this is at the expense of throughput because of the obstruction to the aperture caused by the prism mountings. Other designs use two opposed prisms of differing glasses to provide a direct vision prism (figure 8.23). While this avoids the offset otherwise required to set the telescope onto an object, it reduces the dispersion of the spectrum. The spectral resolution of an objective prism can be theoretically very high because of the large size of the prism (equation 8.33). In practice, however, it will normally be limited by the spatial resolution of the detector because the spectra are physically small due to the low dispersion of the narrow angle prism. On conventional telescopes, objective prisms are being replaced by multi-object instruments (see above) which have higher dispersions and spectral resolutions, can easily provide a wavelength calibration and can use the efficient CCD or IPCS as the detector. Objective prisms and photographic emulsions are still likely to be used for survey purposes for some time to come

Copyright © 1995 IOP Publishing Ltd.

Table 10.10 Parameters of the Objective Prisms for the UK Schmidt Telescope.

Apex angle	Two full aperture prisms with apex angles of 0.73° and 2.25°
Dispersion	The prisms can be used individually or in combination. In the latter case they can be aligned so that their dispersions add together, or they may be opposed to each other and so their dispersions partially cancel. The range of dispersions given by these options varies from 200 to 800 nm mm^{-1} in the red to 47 to 180 nm mm^{-1} in the blue
Spectral resolution	ranges from 37 to 150 in the red and from 40 to 97 in the blue for the various options

on Schmidt cameras, however. The wide field of these instruments means that upwards of a hundred thousand spectra can be obtained in a single exposure, more than offsetting the low efficiency of photography as a detector. The parameters of the objective prisms for the 1.2 m UK Schmidt camera are listed in table 10.10. This camera can also undertake multi-object spectroscopy with images being fed to a conventional laboratory-based spectroscope by fibre optics. This instrument, known as FLAIR (Fibre-Linked Array-Image Reformatter), suffers from a similar constraint to the LDSS (see above) in that an accurately machined plate has to be made for every observation. The fibre optic feeds are mounted onto this plate and it then has to be curved to match the focal surface of the Schmidt camera.

10.14 THE FUTURE

In the absence of some radically new discovery, the future design of spectroscopes seems likely to be a slow evolution towards higher efficiency for the existing types of instrument, and for the use of higher spectral resolutions, at least for some applications. Individual optical components and detectors can already achieve 50% to 80% efficiencies, and so there is only limited scope for improvement there. The overall throughputs for telescope–spectroscope–detector combinations, however, can still be very low. It is here therefore that improvements are likely to arise, especially via a reduction in the number of optical components by the increased use of off-axis surfaces for mirrors and aspherical surfaces for lenses. Variable refractive indices for materials, produced by, for example, the diffusion of silver into glass, are another possible way in which the numbers of components may be reduced. Greater overall efficiency will also be achieved by extending and improving the current trend to multi-object spectroscopy.

Copyright © 1995 IOP Publishing Ltd.

Part 3

Spectroscopy of Astronomical Sources

Copyright © 1995 IOP Publishing Ltd.

11

Spectral Classification

11.1 SPECTRAL CLASS

11.1.1 Core classification

Brief mention of the classification of stars on the basis of the appearance of their spectra has been made in chapter 1, and the historical roots of the rather awkward system that has been evolved discussed there. Here, therefore, we look at the process of spectral classification itself, and the related process of luminosity classification.

The spectral class of a star is based upon the appearance of the lines in its spectrum over the blue–green region, from about 380 nm to 500 nm, and is a classification in terms of the temperature of the stellar surface layers. There are seven major groups in the core of the classification system, labelled, as already mentioned in chapter 1, with the letters

O B A F G K M

in order of decreasing temperature. The mnemonic 'Oh Be A Fine Girl/Guy Kiss Me' may be found useful for remembering the sequence. Each of these major groups is subdivided into ten by the addition of an arabic numeral between 0 and 9. The Sun, with a surface temperature of about 5700 K is thus classified as a G2 star. The hottest stars so far discovered that are classified in this system are O4 stars at about 40 000 K, and the coolest are M8 stars at about 2500 K. Hotter stars exist than O4; white dwarfs in particular may have temperatures up to 100 000 K, but the pressures at their surfaces are so high that few if any spectral lines may be detected (see later discussion and chapter 13), and so they cannot be included within a classification system that is based upon spectral lines. Cooler stars than M8 also almost certainly exist and have been given the rather misleading name of 'Brown Dwarfs'. They are transitional objects between the smallest of the true stars and the largest of the planets. Such objects are the subject of much current observational effort, but at the time of writing

Copyright © 1995 IOP Publishing Ltd.

Table 11.1 The full range of spectral classes in current use.

O4	B0	A0	F0	G0	K0	M0
O5	B0.5	A2	F2	G2	K2	M1
O6	B1	A3	F3	G5	K3	M2
O7	B2	A5	F5	G8	K4	M3
O8	B3	A7	F7		K5	M4
O9	B5		F8			M7
O9.5	B7		F9			M8
	B8					M9
	B9.5					

no unequivocal identification has been made. The relationship between effective surface temperature and spectral class is shown in figure 11.1, for main sequence stars.

More recently the original decimal subdivision of the core classes has been adapted to provide a smoother variation with temperature by adding and deleting some of the classes. Classes have been added between O9 and B0, B0 and B1, and B9 and A0 to give O9.5, B0.5 and B9.5. Classes B4, B6, B9, A1, A4, A6, A8, A9, F1, F4, F6, G1, G3, G4, G6, G7, G9, K1, M5 and M6 have been deleted (K6, K7, K8, K9 were not much used from the beginning); see table 11.1. Some of these classes, however, continue in use in the literature today because the classification in many star catalogues pre-dates the more recent changes.

The appearance of stellar spectra is dominated by a few strong lines which change slowly over the spectral classes (figure 11.2). The principal lines in each class are:

O Mostly highly ionized silicon, nitrogen etc, plus ionized helium (the defining characteristic)

B Neutral helium (the defining characteristic), no ionized helium, lower stages of ionization of silicon and nitrogen, Balmer lines strengthen

A Balmer lines peak at A0, lines from singly ionized calcium and metals appear

F Balmer and ionized metal lines weaken, lines from neutral metals strengthen

G Balmer lines continue to weaken, ionized calcium lines peak in intensity, lines of neutral metals continue to increase in strength

K Many lines due to neutral metals, molecular bands (TiO) start to appear

M TiO bands dominate.

The reason why the lines in a spectrum change with temperature is largely due to the balance between the ionization and the excitation of the atoms composing the outer layers of the star. We have already seen that the population of different excited levels is temperature-dependent, and is given in thermodynamic equilibrium by Boltzmann's formula (equation (4.5)). A useful approximation for much work on stellar atmospheres is that of Local Thermodynamic Equilibrium (LTE). In this approximation, the properties of the material are assumed to be characterized by a single temperature, and the

Copyright © 1995 IOP Publishing Ltd.